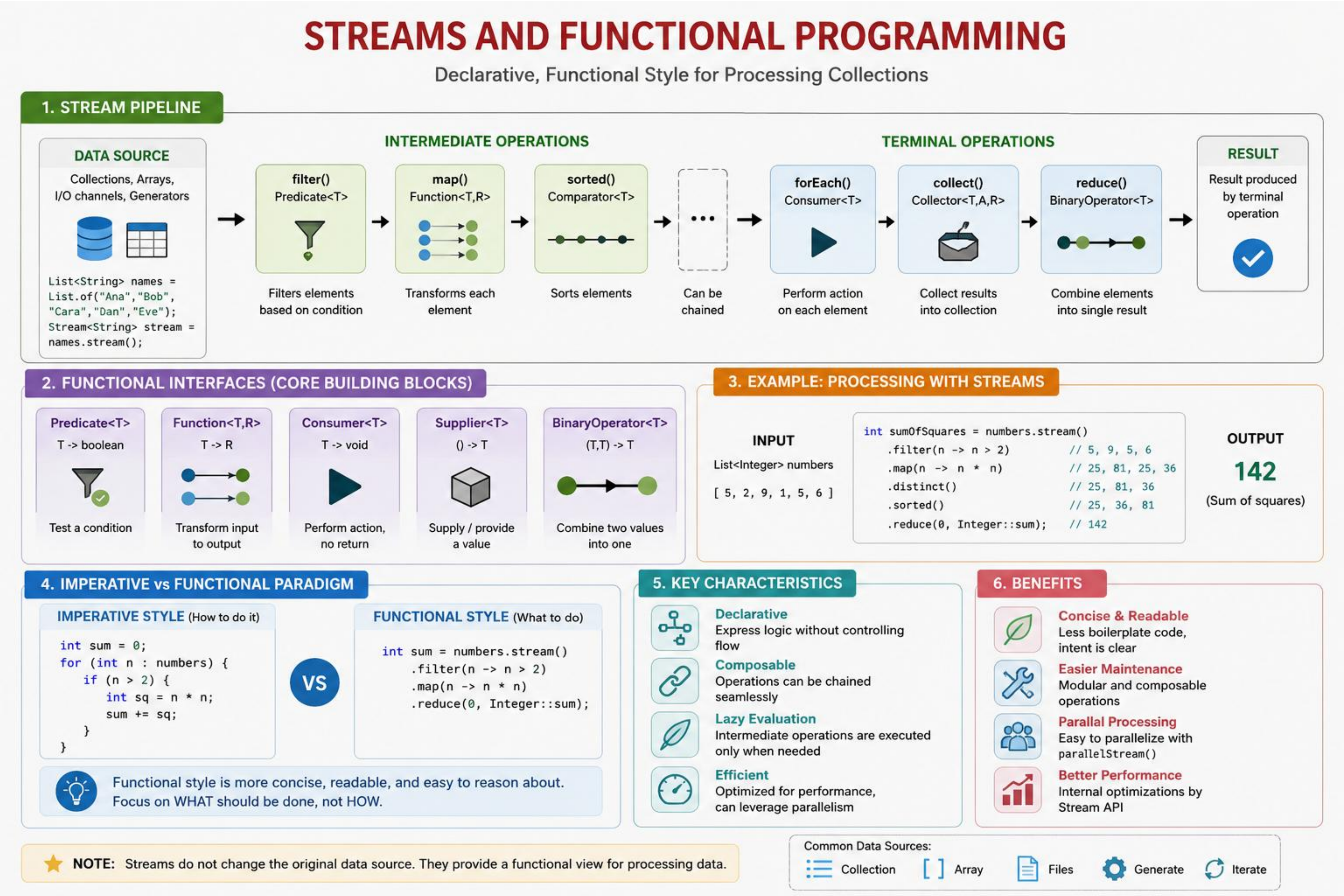
MODULE 6

Streams and Functional Programming

Leverage Java Streams and functional programming to write concise, expressive, and efficient code.

λ 🔗 ⏳ 📦 ⚖️


TEKsystems
Global Services
WORKFORCE
DEVELOPMENT


STREAMS AND FUNCTIONAL PROGRAMMING
Declarative, Functional Style for Processing Collections
1. STREAM PIPELINE
DATA SOURCE
Collections, Arrays, I/O channels, Generators
List<String> names = List.of("Ana","Bob", "Cara","Dan","Eve"); Stream<String> stream = names.stream();
INTERMEDIATE OPERATIONS
filter() Predicate<T>
Filters elements based on condition
map() Function<T,R>
Transforms each element
sorted() Comparator<T>
Sorts elements
...
Can be chained
TERMINAL OPERATIONS
forEach() Consumer<T>
Perform action on each element
collect() Collector<T,A,R>
Collect results into collection
reduce() BinaryOperator<T>
Combine elements into single result
RESULT
Result produced by terminal operation
2. FUNCTIONAL INTERFACES (CORE BUILDING BLOCKS)
Predicate<T>
T -> boolean
Test a condition
Function<T,R>
T -> R
Transform input to output
Consumer<T>
T -> void
Perform action, no return
Supplier<T>
() -> T
Supply / provide a value
BinaryOperator<T>
(T,T) -> T
Combine two values into one
3. EXAMPLE: PROCESSING WITH STREAMS
INPUT
List<Integer> numbers
[5, 2, 9, 1, 5, 6]
int sumOfSquares = numbers.stream()
.filter(n -> n > 2) // 5, 9, 5, 6
.map(n -> n * n) // 25, 81, 25, 36
.distinct() // 25, 81, 36
.sorted() // 25, 36, 81
.reduce(0, Integer::sum); // 142
OUTPUT
142
(Sum of squares)
4. IMPERATIVE vs FUNCTIONAL PARADIGM
IMPERATIVE STYLE (How to do it)
int sum = 0;
for (int n : numbers) {
if (n > 2) {
int sq = n * n;
sum += sq;
}
}
VS
FUNCTIONAL STYLE (What to do)
int sum = numbers.stream()
.filter(n -> n > 2)
.map(n -> n * n)
.reduce(0, Integer::sum);
Functional style is more concise, readable, and easy to reason about.
Focus on WHAT should be done, not HOW.
5. KEY CHARACTERISTICS
Declarative
Express logic without controlling flow
Composable
Operations can be chained seamlessly
Lazy Evaluation
Intermediate operations are executed only when needed
Efficient
Optimized for performance, can leverage parallelism
6. BENEFITS
Concise & Readable
Less boilerplate code, intent is clear
Easier Maintenance
Modular and composable operations
Parallel Processing
Easy to parallelize with parallelStream()
Better Performance
Internal optimizations by Stream API
NOTE: Streams do not change the original data source. They provide a functional view for processing data.
Common Data Sources:
Collection
[ ] Array
Files
Generate
Iterate

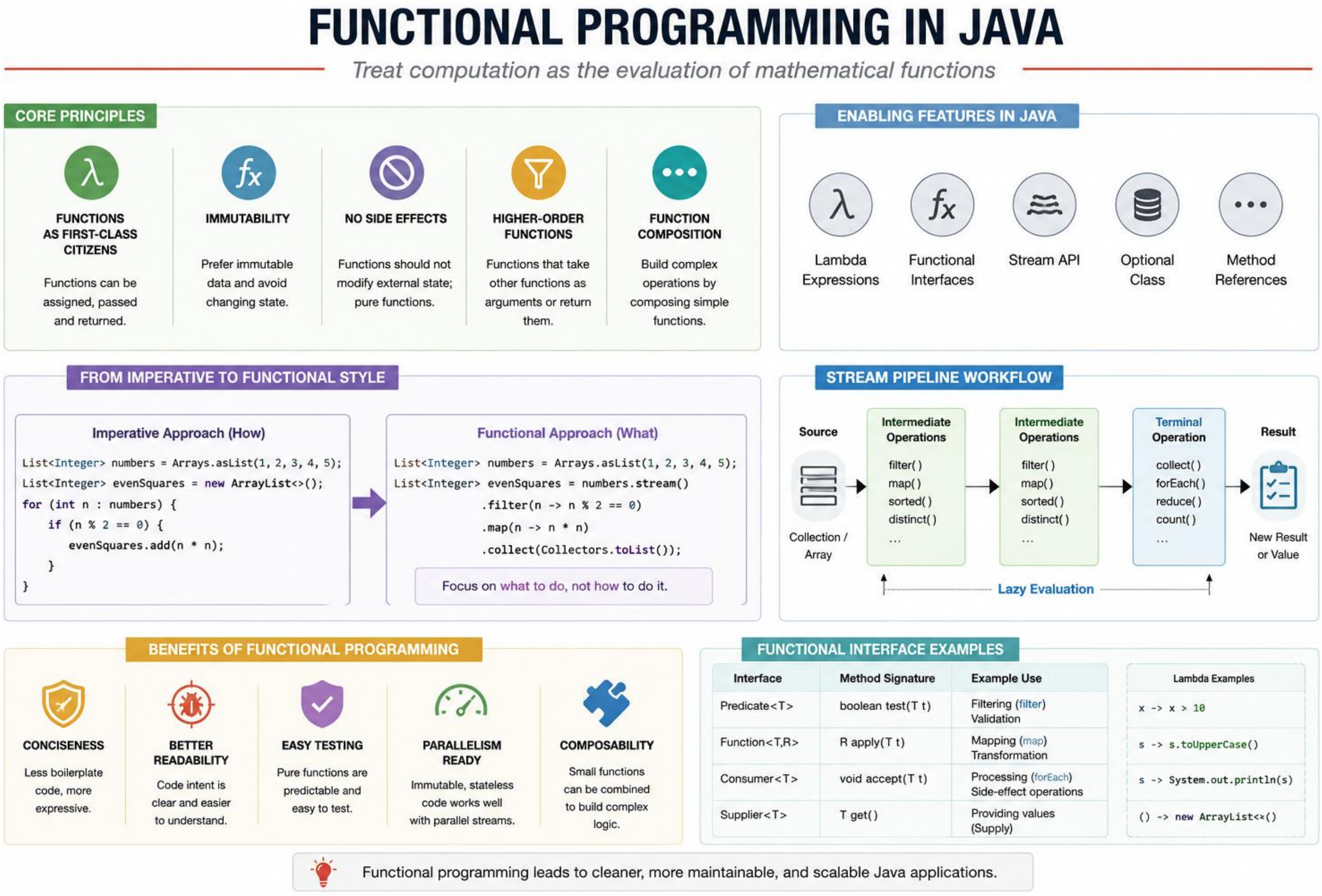
MODULE OVERVIEW

Leverage Java Streams and functional programming features to write concise, expressive, and efficient code.

WHAT YOU WILL LEARN

- Introduction to Streams and the Stream Pipeline
- Creating Streams from Collections, Arrays, and other sources
- Intermediate Operations: filter, map, flatMap, sorted, distinct, peek, limit, skip
- Terminal Operations: forEach, collect, reduce, count, anyMatch, allMatch, noneMatch, findFirst
- Stream Pipeline Concepts: lazy evaluation, short-circuiting, stateless vs stateful operations
- Functional Interfaces: Predicate, Function, Consumer, Supplier
- Lambda Expressions and Method References
- Optional class and handling absence of values
- Best Practices and Performance Considerations

KEY TAKEAWAY Streams + Functional Programming enable declarative, readable, and efficient data processing while reducing boilerplate.



ALSO FROM THIS TOPIC Less error-prone — immutable data & pure functions reduce bugs • Designed for modern systems — aligns with streams, reactive & cloud apps

IMPERATIVE VS FUNCTIONAL APPROACH

The same task, expressed two different ways.

TRADITIONAL IMPERATIVE APPROACH

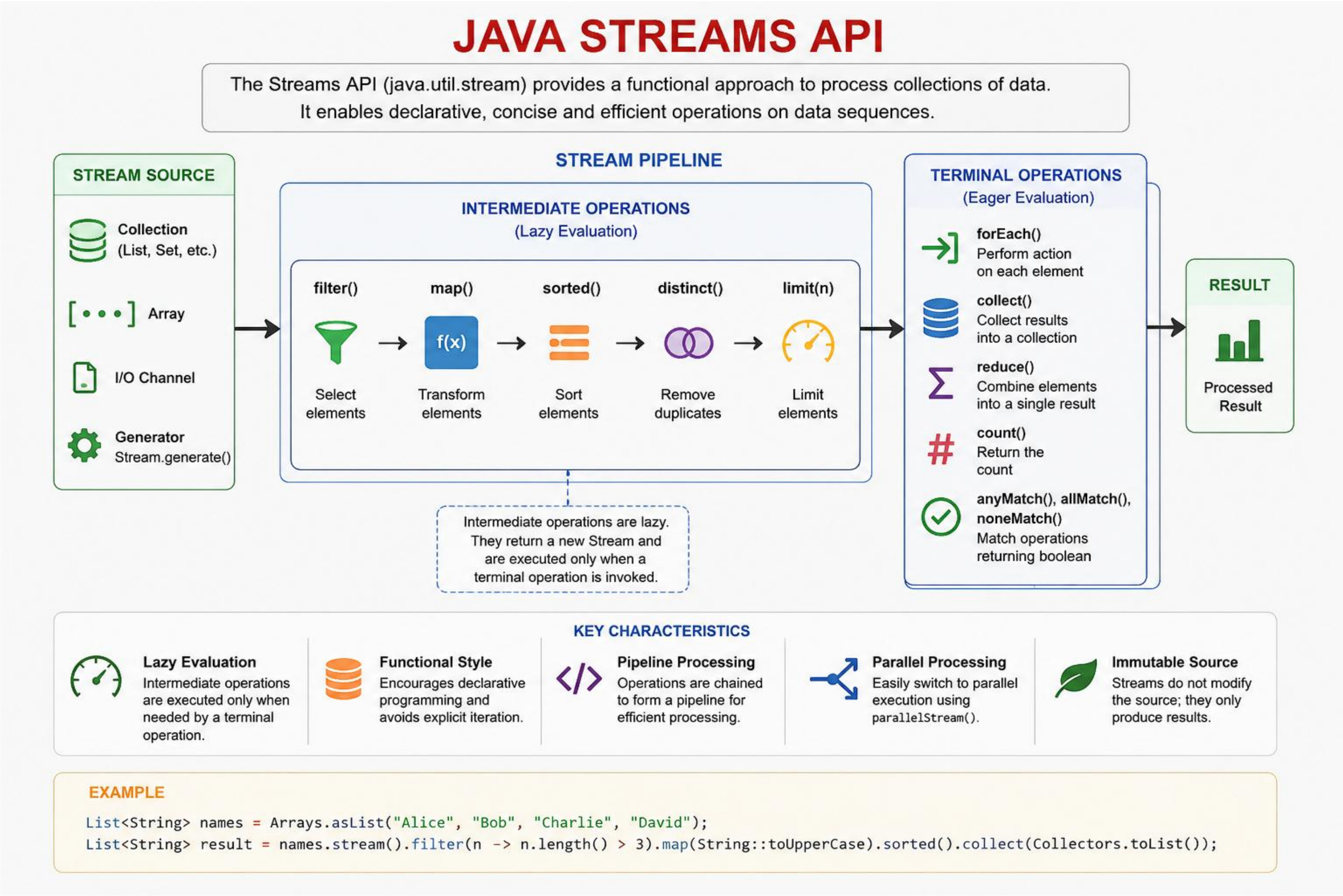
```
List<Integer> result = new ArrayList<>();
for (int n : numbers) {
    if (n % 2 == 0) {
        result.add(n * 2);
    }
}
```

FUNCTIONAL APPROACH WITH STREAMS

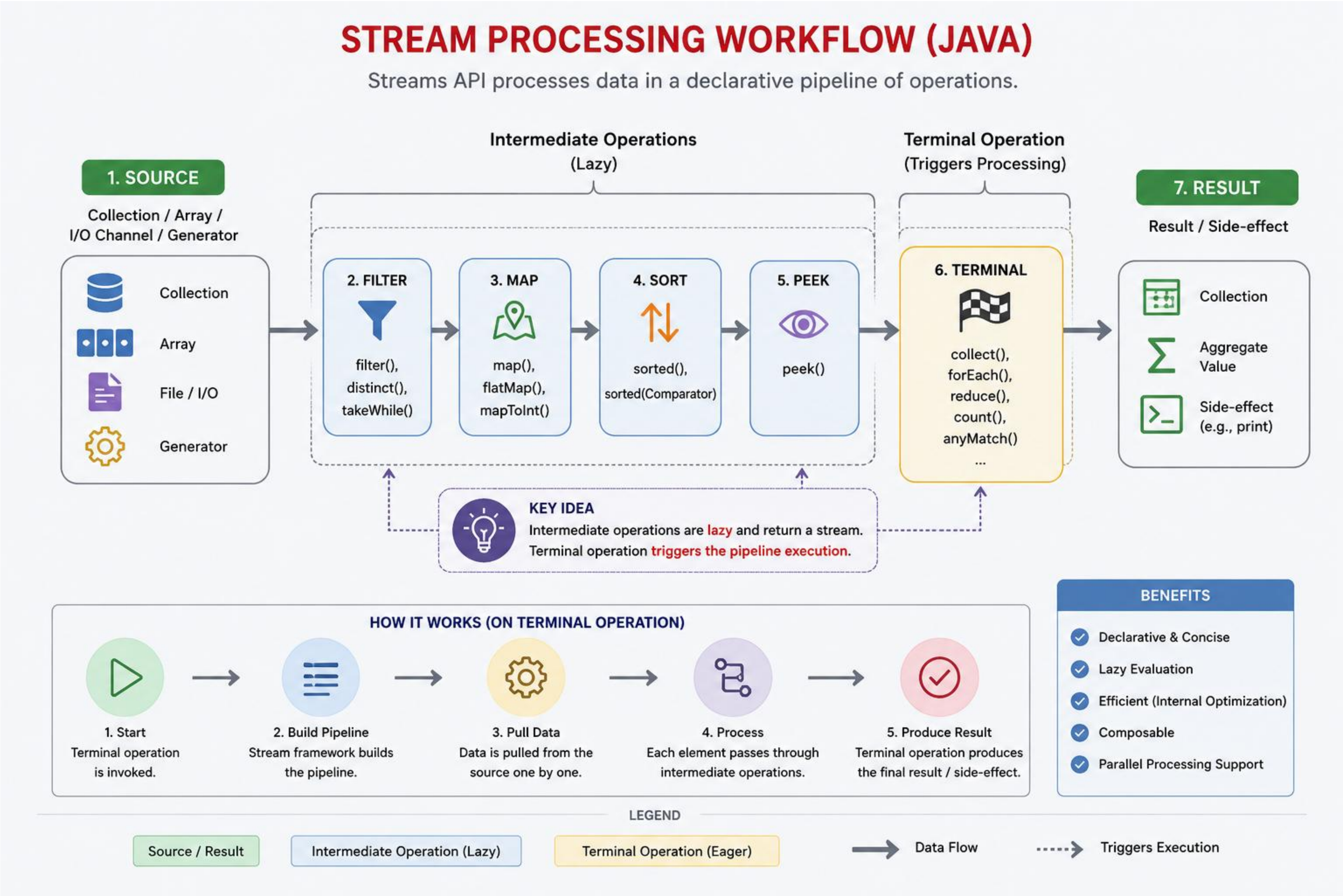
```
List<Integer> result = numbers.stream()
    .filter(n -> n % 2 == 0)
    .map(n -> n * 2)
    .collect(Collectors.toList());
```

- Imperative: more code, more state, more room for errors.
- Functional: less code, easier to read, safer, and more expressive.

KEY TAKEAWAY Functional programming is not just a style — it leads to robust, scalable, future-ready software.

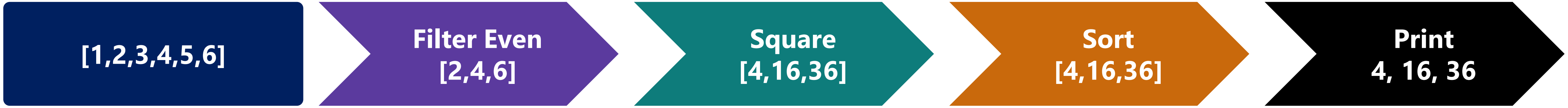


ALSO FROM THIS TOPIC Single-use — a stream can be consumed only once • Not a data structure — no add/remove/update operations


STREAM PROCESSING WORKFLOW (JAVA)
Streams API processes data in a declarative pipeline of operations.
Intermediate Operations (Lazy)
Terminal Operation (Triggers Processing)
1. SOURCE
Collection / Array / I/O Channel / Generator
Collection
Array
File / I/O
Generator
2. FILTER
filter(), distinct(), takeWhile()
3. MAP
map(), flatMap(), mapToInt()
4. SORT
sorted(), sorted(Comparator)
5. PEEK
peek()
6. TERMINAL
collect(), forEach(), reduce(), count(), anyMatch() ...
7. RESULT
Result / Side-effect
Collection
Aggregate Value
Side-effect (e.g., print)
KEY IDEA
Intermediate operations are lazy and return a stream.
Terminal operation triggers the pipeline execution.
HOW IT WORKS (ON TERMINAL OPERATION)
1. Start
Terminal operation is invoked.
2. Build Pipeline
Stream framework builds the pipeline.
3. Pull Data
Data is pulled from the source one by one.
4. Process
Each element passes through intermediate operations.
5. Produce Result
Terminal operation produces the final result / side-effect.
BENEFITS
Declarative & Concise
Lazy Evaluation
Efficient (Internal Optimization)
Composable
Parallel Processing Support
LEGEND
Source / Result
Intermediate Operation (Lazy)
Terminal Operation (Eager)
Data Flow
Triggers Execution

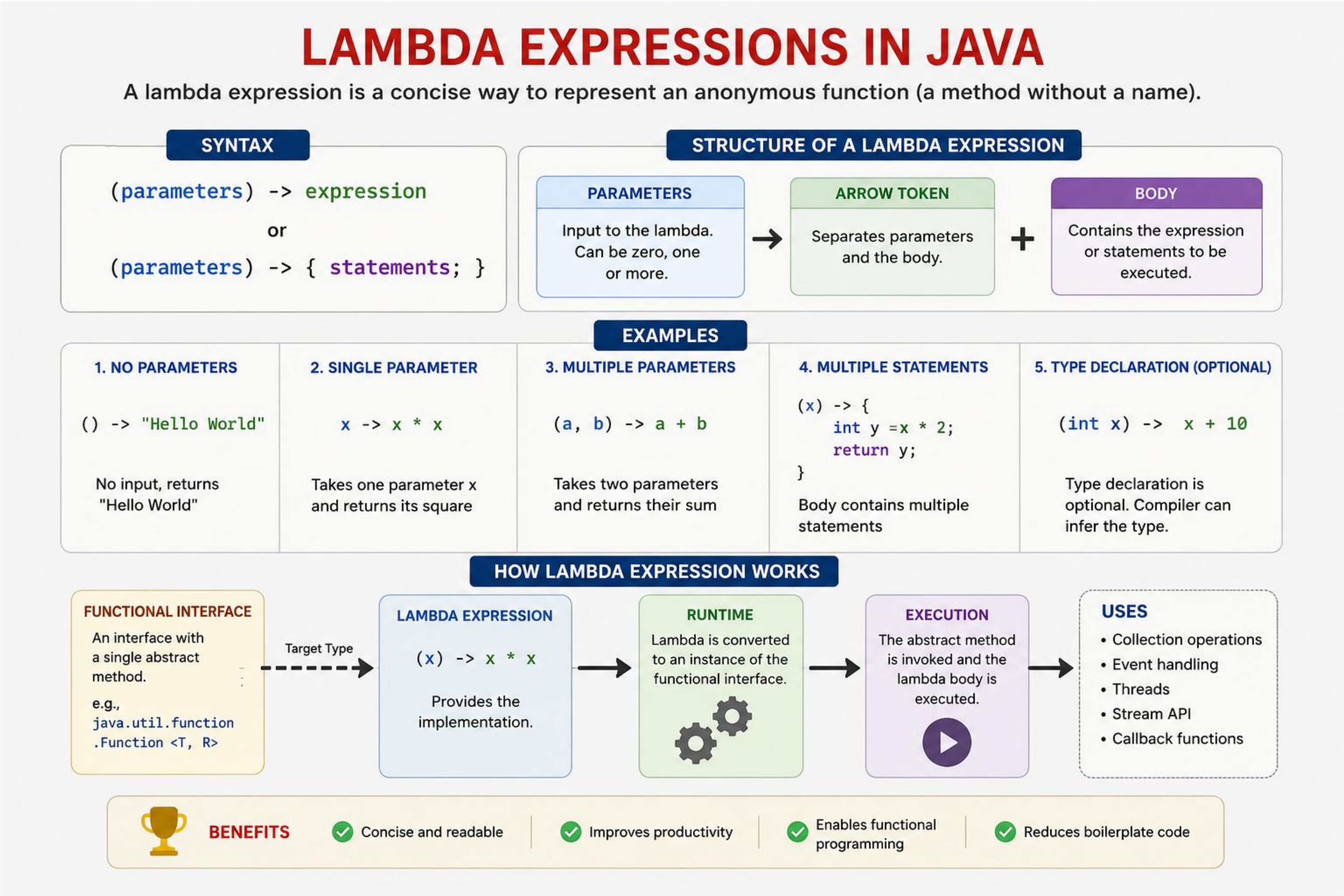
EXAMPLE SCENARIO

From a list of numbers: get the evens, square them, sort, and print.




```java
List<Integer> numbers = Arrays.asList(1, 2, 3, 4, 5, 6);
numbers.stream()
    .filter(n -> n % 2 == 0)       // intermediate
    .map(n -> n * n)               // intermediate
    .sorted()                      // intermediate
    .forEach(System.out::println); // terminal
```

KEY TAKEAWAY Every intermediate step in this diagram is lazy — only forEach() actually runs the pipeline.


LAMBDA EXPRESSIONS IN JAVA
A lambda expression is a concise way to represent an anonymous function (a method without a name).
SYNTAX
(parameters) -> expression
or
(parameters) -> { statements; }
STRUCTURE OF A LAMBDA EXPRESSION
PARAMETERS
Input to the lambda. Can be zero, one or more.
ARROW TOKEN
Separates parameters and the body.
BODY
Contains the expression or statements to be executed.
EXAMPLES
1. NO PARAMETERS
() -> "Hello World"
No input, returns "Hello World"
2. SINGLE PARAMETER
x -> x * x
Takes one parameter x and returns its square
3. MULTIPLE PARAMETERS
(a, b) -> a + b
Takes two parameters and returns their sum
4. MULTIPLE STATEMENTS
(x) -> {
int y =x * 2;
return y;
}
Body contains multiple statements
5. TYPE DECLARATION (OPTIONAL)
(int x) -> x + 10
Type declaration is optional. Compiler can infer the type.
HOW LAMBDA EXPRESSION WORKS
FUNCTIONAL INTERFACE
An interface with a single abstract method.
e.g., java.util.function .Function <T, R>
Target Type
LAMBDA EXPRESSION
(x) -> x * x
Provides the implementation.
RUNTIME
Lambda is converted to an instance of the functional interface.
EXECUTION
The abstract method is invoked and the lambda body is executed.
USES
• Collection operations
• Event handling
• Threads
• Stream API
• Callback functions
BENEFITS
Concise and readable
Improves productivity
Enables functional programming
Reduces boilerplate code

LAMBDA VS ANONYMOUS CLASS

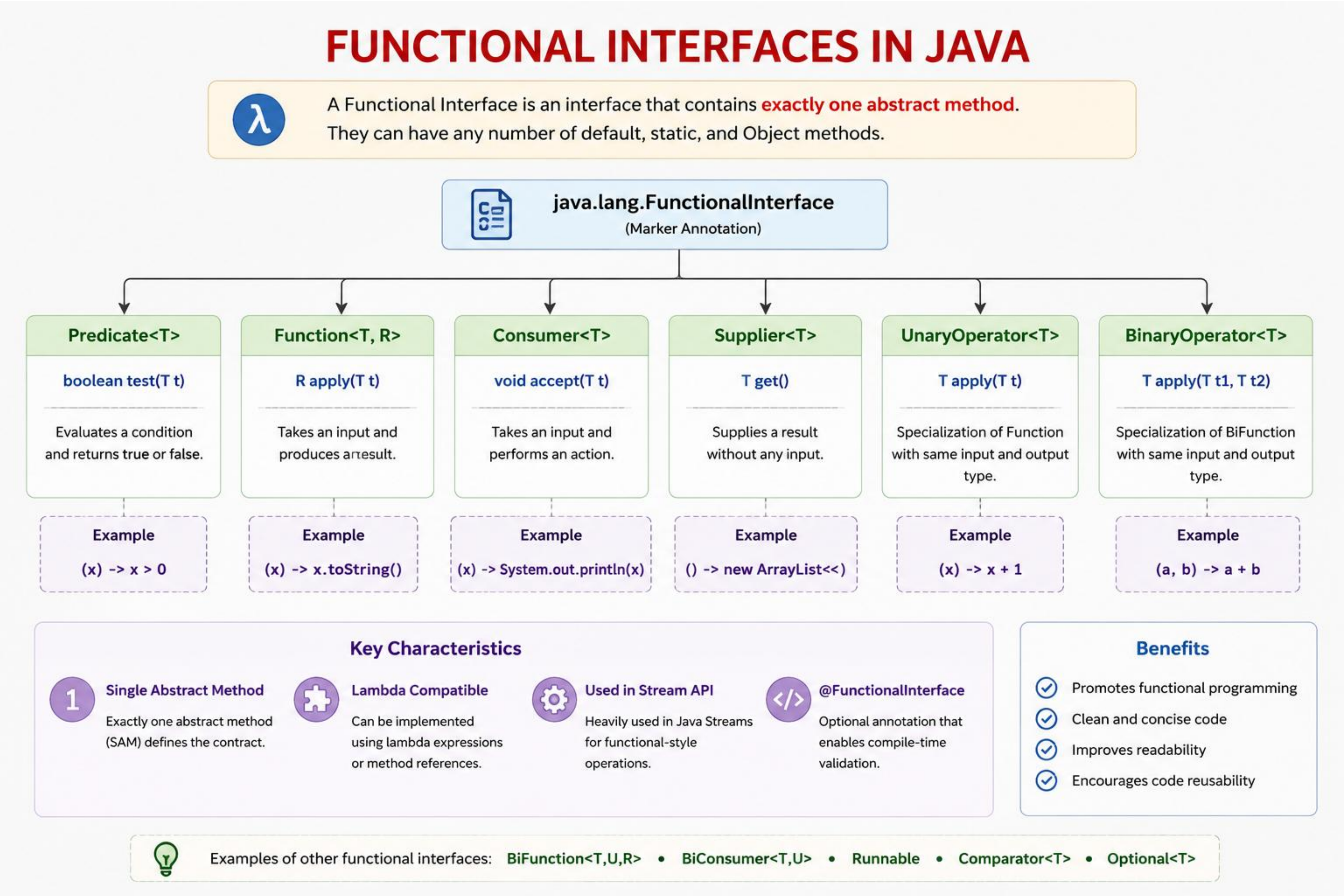
The same behavior, five different shapes — lambdas are always shorter.

SCENARIO	LAMBDA	EQUIVALENT ANONYMOUS CLASS
`No params, return a value`	() -> 42	new Supplier<Integer>() {...}
`One param, return a value`	x -> x * x	new Function<Integer,Integer>() {...}
`Multiple params, return a value`	(a, b) -> a + b	new BiFunction<...>() {...}
`No params, no return (void)`	() -> System.out.println("Hi")	new Runnable() {...}
`One param, no return (void)`	msg -> System.out.println(msg)	new Consumer<String>() {...}

WHERE LAMBDAS ARE USED Collection processing, event handling, executors, custom APIs — anywhere Java expects a functional interface.

```
names.forEach(name -> System.out.println(name.toUpperCase()));  // ALICE BOB CHARLIE
```

KEY TAKEAWAY Every lambda on the left compiles down to the same kind of object shown on the right — just with far less ceremony.


FUNCTIONAL INTERFACES IN JAVA
λ
A Functional Interface is an interface that contains exactly one abstract method.
They can have any number of default, static, and Object methods.
java.lang.FunctionalInterface
(Marker Annotation)
Predicate<T>
boolean test(T t)
Evaluates a condition and returns true or false.
Example
(x) -> x > 0
Function<T, R>
R apply(T t)
Takes an input and produces a result.
Example
(x) -> x.toString()
Consumer<T>
void accept(T t)
Takes an input and performs an action.
Example
(x) -> System.out.println(x)
Supplier<T>
T get()
Supplies a result without any input.
Example
() -> new ArrayList<<)
UnaryOperator<T>
T apply(T t)
Specialization of Function with same input and output type.
Example
(x) -> x + 1
BinaryOperator<T>
T apply(T t1, T t2)
Specialization of BiFunction with same input and output type.
Example
(a, b) -> a + b
Key Characteristics
1
Single Abstract Method
Exactly one abstract method (SAM) defines the contract.
Lambda Compatible
Can be implemented using lambda expressions or method references.
Used in Stream API
Heavily used in Java Streams for functional-style operations.
</>
@FunctionalInterface
Optional annotation that enables compile-time validation.
Benefits
Promotes functional programming
Clean and concise code
Improves readability
Encourages code reusability
Examples of other functional interfaces: BiFunction<T,U,R> • BiConsumer<T,U> • Runnable • Comparator<T> • Optional<T>

WHY THEY MATTER & COMMON INTERFACES

The foundation that lets you pass behavior as data — enabling lambdas, method references, and the Streams API.

INTERFACE	TYPE	ABSTRACT METHOD	DESCRIPTION
`Predicate<T>`	Function	boolean test(T t)	Evaluates a condition.
`Function<T,R>`	Function	R apply(T t)	Transforms input to output.
`BiFunction<T,U,R>`	Function	R apply(T t, U u)	Works on two inputs, produces a result.
`Consumer<T>`	Consumer	void accept(T t)	Consumes a value, returns no result.
`Supplier<T>`	Supplier	T get()	Supplies a value, takes no input.
`BiPredicate<T,U>`	Function	boolean test(T t, U u)	Evaluates a condition on two inputs.

KEY TAKEAWAY Predicate, Function, Consumer, and Supplier cover almost every functional-interface need in everyday code.

USING A FUNCTIONAL INTERFACE WITH A LAMBDA

The lambda supplies the implementation of the single abstract method.

```
MyFunctionalInterface obj =
    (message) -> System.out.println("Hello " + message);

obj.perform("Java");   // Output: Hello Java
```

Functional interfaces are the foundation of Java's functional programming. They allow behavior to be passed as an argument, enabling powerful and expressive code with lambdas, method references, and the Streams API.

Declaring your own: `@FunctionalInterface public interface Name { returnType method(params); }`

Default and static methods are optional extras — only the single abstract method counts toward the rule.

KEY TAKEAWAY The lambda expression provides the implementation for the abstract method defined in the functional interface.

TRY IT YOURSELF — EXERCISE 1: LAMBDA EXPRESSIONS

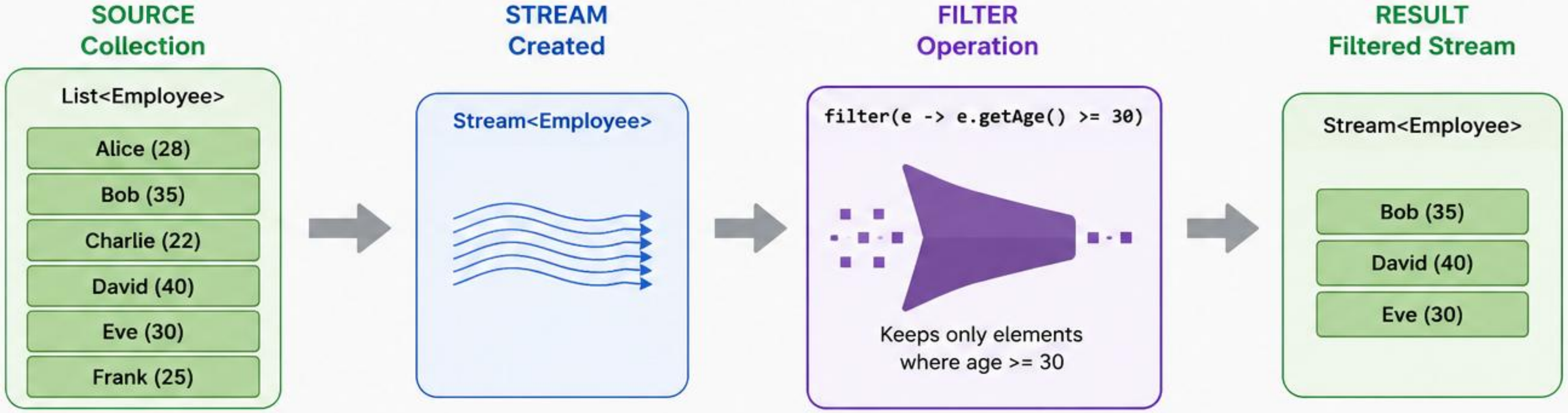
Reinforces: lambdas and functional interfaces, as just covered.

STEP	CODE	RESULT
`Goal`	Implement a one-method interface as an anonymous class and as a lambda	Both produce the same result
`Code`	SalaryCheck lambda = e -> e.salary() > 60_000;	A lambda implementing SalaryCheck.test(Employee)
`Key concept`	A functional interface has exactly one abstract method	That single method is what a lambda implements
`Reference`	exercises/exercise-01-lambda-functional-interface.md	Full starter code and steps

TRY IT NOW Complete Exercise 1 before continuing — see exercises/exercise-01-lambda-functional-interface.md.

FILTERING DATA WITH STREAMS (JAVA)

The `filter()` intermediate operation selects elements that match a given **predicate (condition)** and returns a new Stream containing only those elements.



EXAMPLE CODE

```java
List<Employee> employees = Arrays.asList(
    new Employee("Alice", 28),
    new Employee("Bob", 35),
    new Employee("Charlie", 22),
    new Employee("David", 40),
    new Employee("Eve", 30),
    new Employee("Frank", 25)
);
List<Employee> filtered = employees.stream()
    .filter(e -> e.getAge() >= 30)    // Filtering condition
    .collect(Collectors.toList());    // Collect results
```

KEY POINTS

- `filter()` is an intermediate operation.
- Takes a `Predicate<T>` and returns a `Stream<T>`.
- Does not modify the original data.
- Supports method references and lambda expressions.

PREDICATE EXAMPLES

Using Lambda Expression	Using Method Reference (if applicable)	Compound Condition
`.filter(e -> e.getAge() >= 30)`	`.filter(Employee::isActive)` `// where isActive() returns boolean`	`.filter(e -> e.getAge() >= 30 && e.getSalary() > 50000)`

REAL-WORLD EXAMPLE: FILTER EMPLOYEES BY SALARY

Keep only employees earning more than 60,000.

NAME	DEPARTMENT	SALARY
Alice	IT	85000
Bob	HR	45000
Charlie	IT	95000
David	Finance	55000
Eve	IT	90000

```
employees.stream()
  .filter(e ->
    e.getSalary() > 60000)
  .collect(
    Collectors.toList());

// Alice, Charlie, Eve
```

KEY TAKEAWAY Use filter() to narrow your data down to only the elements that match a condition — the foundation of stream processing.

TRY IT YOURSELF — EXERCISE 2: FILTER BY SALARY

Reinforces: the filter operation you just saw.

STEP	CODE	RESULT
`Goal`	List employees with salary greater than 60,000	Alice, Bob, Charlie, Diana (not Evan)
`Code`	.filter(e -> e.salary() > 60_000)	A shorter stream — Evan is excluded
`Key concept`	filter() keeps only elements matching a boolean predicate	Ties to the filtering operations you just saw
`Reference`	exercises/exercise-02-filter-salary.md	Full starter code and steps

TRY IT NOW Complete Exercise 2 before continuing — see exercises/exercise-02-filter-salary.md.

MAPPING DATA WITH STREAMS

Mapping transforms each element of a stream to another object using the `map()` or `flatMap()` operations.

SOURCE DATA

List<Employee> employees

- Employee(1, "Alice", 70000)
- Employee(2, "Bob", 60000)
- Employee(3, "Charlie", 80000)
- Employee(4, "Diana", 65000)

STREAM PIPELINE (MAPPING)

map() — Transform each Employee to Employee Name → **map()** — Transform each Name to Uppercase → **forEach()** — Consume the results

```java
employees.stream()
    .map(emp -> emp.getName())      // Employee → Name
    .map(String::toUpperCase)       // Name → Uppercase
    .forEach(System.out::println); // Print result
```

MAPPED OUTPUT

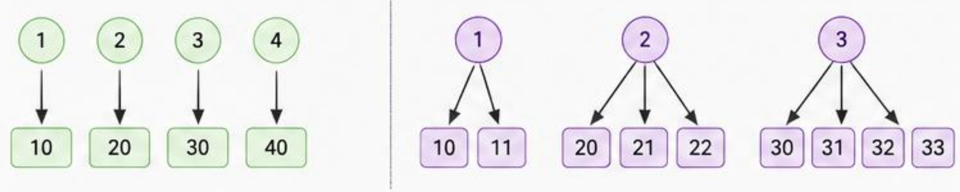
Stream<String> (Uppercase Names)

- ALICE
- BOB
- CHARLIE
- DIANA

map() vs flatMap()

map() One input → One output

flatMap() One input → 0..N outputs



Use map() for simple transformations, flatMap() for nested collections or expanding data.

COMMON MAPPING EXAMPLES

Goal	Example
Extract specific field	employees.stream().map(Employee::getName)
Convert to another type	employees.stream().map(Employee::getId)
Compute derived value	employees.stream().map(emp -> emp.getSalary() * 1.1)
Flatten nested collections	departments.stream() .flatMap(dept -> dept.getEmployees().stream())

BEST PRACTICES

- Use map() for one-to-one transformation.
- Use flatMap() to flatten nested streams or collections.
- Keep mapping functions stateless and side-effect free.
- Chain multiple map() operations for complex transformations.

EXAMPLE: GET LENGTH OF EACH STRING

A second mapping example, plus where map() shows up in practice.

```
List<Integer> lengths = names.stream()
    .map(name -> name.length())   // length of each string
    .collect(Collectors.toList());  // [5, 3, 7]
```

- Common use cases: data transformation, format conversion, extracting properties,
- calculations, and preparing data for other operations in the pipeline.

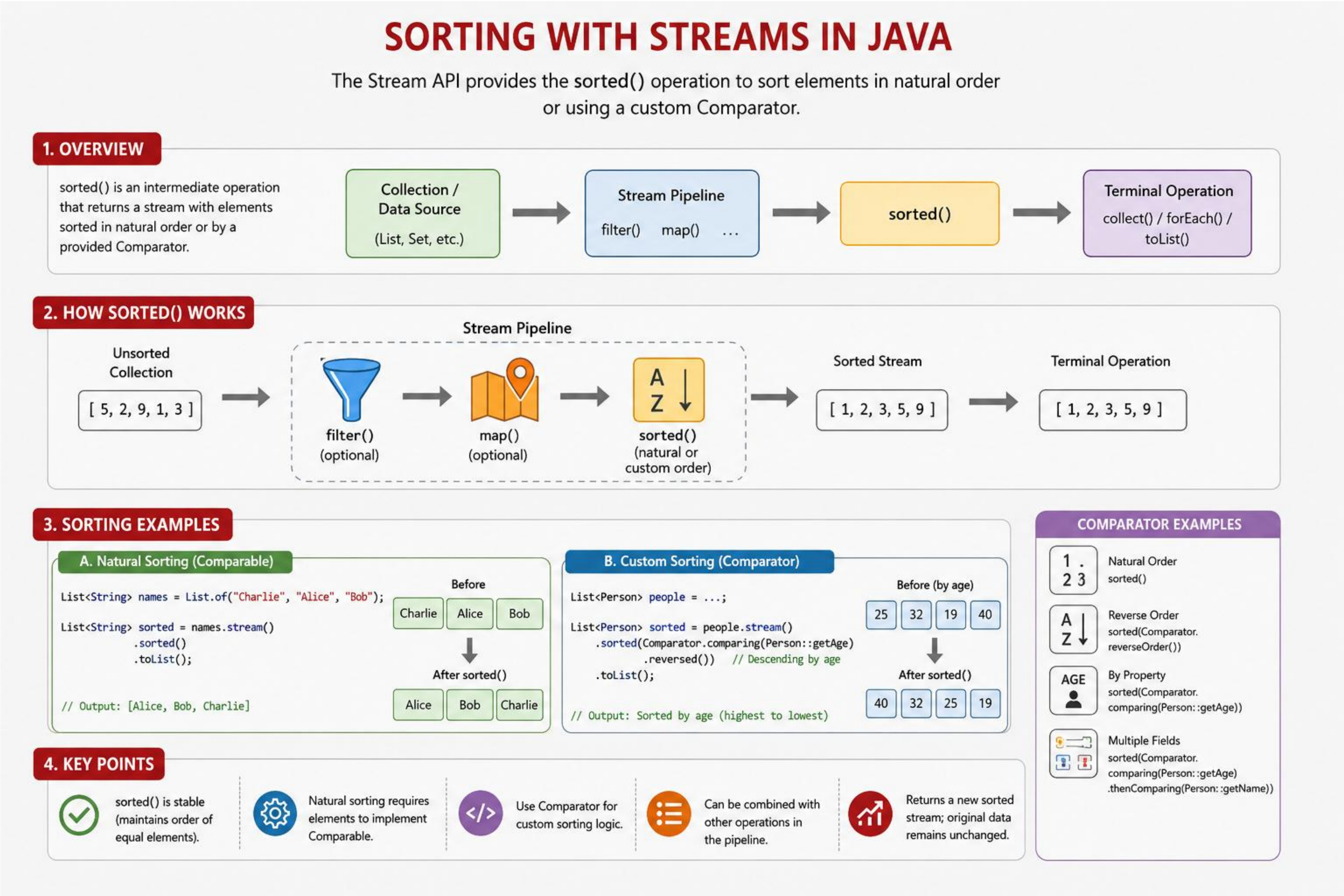
KEY TAKEAWAY Use map() to transform each element into a new form — a powerful tool for shaping data for the next pipeline step.

TRY IT YOURSELF — EXERCISE 3: LIST ALL NAMES

Reinforces: the map operation you just saw.

STEP	CODE	RESULT
`Goal`	Print all employee names using map + forEach or collect	All five names print
`Code`	.map(Employee::name).forEach(System.out::println)	Alice, Bob, Charlie, Diana, Evan
`Key concept`	map() transforms each element — Employee to String here	Ties to the mapping operations you just saw
`Reference`	exercises/exercise-03-list-names.md	Full starter code and steps

TRY IT NOW Complete Exercise 3 before continuing — see exercises/exercise-03-list-names.md.



SORTING WITH STREAMS IN JAVA
The Stream API provides the sorted() operation to sort elements in natural order or using a custom Comparator.
1. OVERVIEW
sorted() is an intermediate operation that returns a stream with elements sorted in natural order or by a provided Comparator.
Collection / Data Source
(List, Set, etc.)
Stream Pipeline
filter() map() ...
sorted()
Terminal Operation
collect() / forEach() / toList()
2. HOW SORTED() WORKS
Stream Pipeline
Unsorted Collection
[5, 2, 9, 1, 3]
filter()
(optional)
map()
(optional)
A Z
sorted()
(natural or custom order)
Sorted Stream
[1, 2, 3, 5, 9]
Terminal Operation
[1, 2, 3, 5, 9]
3. SORTING EXAMPLES
A. Natural Sorting (Comparable)
List<String> names = List.of("Charlie", "Alice", "Bob");
List<String> sorted = names.stream()
.sorted()
.toList();
// Output: [Alice, Bob, Charlie]
Before
Charlie Alice Bob
After sorted()
Alice Bob Charlie
B. Custom Sorting (Comparator)
List<Person> people = ...;
List<Person> sorted = people.stream()
.sorted(Comparator.comparing(Person::getAge)
.reversed()) // Descending by age
.toList();
// Output: Sorted by age (highest to lowest)
Before (by age)
25 32 19 40
After sorted()
40 32 25 19
COMPARATOR EXAMPLES
Natural Order
sorted()
Reverse Order
sorted(Comparator.reverseOrder())
AGE
By Property
sorted(Comparator.comparing(Person::getAge))
Multiple Fields
sorted(Comparator.comparing(Person::getAge).thenComparing(Person::getName))
4. KEY POINTS
sorted() is stable (maintains order of equal elements).
Natural sorting requires elements to implement Comparable.
Use Comparator for custom sorting logic.
Can be combined with other operations in the pipeline.
Returns a new sorted stream; original data remains unchanged.

EXAMPLE: SORT OBJECTS BY SALARY (DESCENDING)

Comparator.comparing(...).reversed() sorts custom objects in descending order.

```
List<Employee> sortedBySalary = employees.stream()
    .sorted(Comparator.comparing(Employee::getSalary)
        .reversed())        // descending by salary
    .collect(Collectors.toList());
```

NAME	SALARY (DESC)
Bob	95000
Alice	75000
Charlie	55000

MORE SORTING PATTERNS

sorted() — natural order

sorted(Comparator.reverseOrder()) — reverse order

sorted(String.CASE_INSENSITIVE_ORDER) — case-insensitive

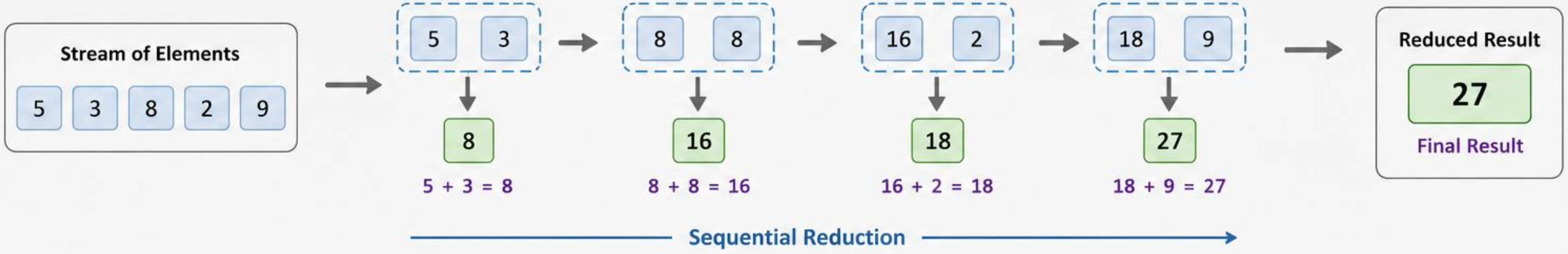
sorted(Comparator.comparing(p -> p.name())) — by property

KEY TAKEAWAY Use sorted() with natural ordering or a custom Comparator for flexible, expressive sorting.

REDUCTION OPERATIONS IN JAVA

Reduction operations combine the elements of a stream into a single result using an identity value and an associative accumulation function.

HOW REDUCTION WORKS



TYPES OF REDUCTION (Stream API)

1. **reduce(identity, accumulator)**
   Combines elements with an identity value.
   Returns identity if stream is empty.

```
Optional<T> reduce(T identity,
              BinaryOperator<T> accumulator)
```

2. **reduce(accumulator)**
   Combines elements without an identity.
   Returns `Optional.empty()` if stream is empty.

```
Optional<T> reduce(BinaryOperator<T> accumulator)
```

3. **reduce(identity, accumulator, combiner)**
   Used in parallel streams. Combines partial results using combiner.

```
T reduce(T identity, BinaryOperator<T> accumulator,
           BinaryOperator<T> combiner)
```

COMMON EXAMPLES

		Result
Σ	**Sum** `reduce(0, Integer::sum)`	27
×	**Product** `reduce(1, (a, b) -> a * b)`	2160
max	**Maximum** `reduce(Integer::max)`	9
min	**Minimum** `reduce(Integer::min)`	2
concat	**Concatenate Strings** `reduce("", (a, b) -> a + b)`	"53829"

CODE EXAMPLE

```
List<Integer> numbers = Arrays.asList(5, 3, 8, 2, 9);

// 1. With identity
int sum = numbers.stream()
        .reduce(0, Integer::sum);
System.out.println("Sum = " + sum);  // 27

// 2. Without identity
Optional<Integer> max = numbers.stream()
        .reduce(Integer::max);
max.ifPresent(m -> System.out.println("Max = " + m)); // 9

// 3. Parallel stream with combiner
int sumParallel = numbers.parallelStream()
        .reduce(0, Integer::sum, Integer::sum);
System.out.println("Parallel Sum = " + sumParallel); // 27
```

KEY POINTS

- Reduction produces a single summary result from the stream.
- The accumulator function must be associative.
- Use identity when a default starting value exists.
- For parallel streams, provide a proper combiner function.

Works with Sequential & Parallel Streams

REDUCTION METHODS OVERVIEW

Eight methods, each with a distinct return type.

METHOD	RETURN TYPE	DESCRIPTION	EXAMPLE
`reduce(identity, accumulator)`	T	Combines using identity + accumulator	reduce(0, Integer::sum)
`reduce(accumulator)`	Optional<T>	Combines without an identity value	reduce(Integer::sum)
`sum()`	int/long/double	Sum of numeric stream elements	mapToInt(i->i).sum()
`min(comparator) / max(comparator)`	Optional<T>	Smallest / largest element	min(Comparator.naturalOrder())
`count()`	long	Number of elements	count()
`average()`	OptionalDouble	Average of numeric elements	mapToInt(i->i).average()
`collect(collector)`	R	Reduces using a Collector	collect(Collectors.toList())

KEY TAKEAWAY Know your return types: Optional<T>, OptionalDouble, long, int — each reduction returns a different shape.

REDUCTION — CODE EXAMPLES

reduce, min, max, count, and average, all on the same data.

```
List<Integer> numbers = Arrays.asList(2, 4, 6, 8, 10);

int sum = numbers.stream().reduce(0, Integer::sum);          // 30
Optional<Integer> sumOpt = numbers.stream().reduce(Integer::sum);  // Optional[30]
Optional<Integer> min = numbers.stream().min(Integer::compare);    // Optional[2]
Optional<Integer> max = numbers.stream().max(Integer::compare);    // Optional[10]
long count = numbers.stream().count();                             // 5
OptionalDouble avg = numbers.stream()
    .mapToInt(Integer::intValue).average();  // OptionalDouble[6.0]

// Custom reduction (no built-in method covers this):
String csv = names.stream().reduce("", (a, b) -> a.isEmpty() ? b : a + ", " + b);  // Alice, Bob, Charlie
```

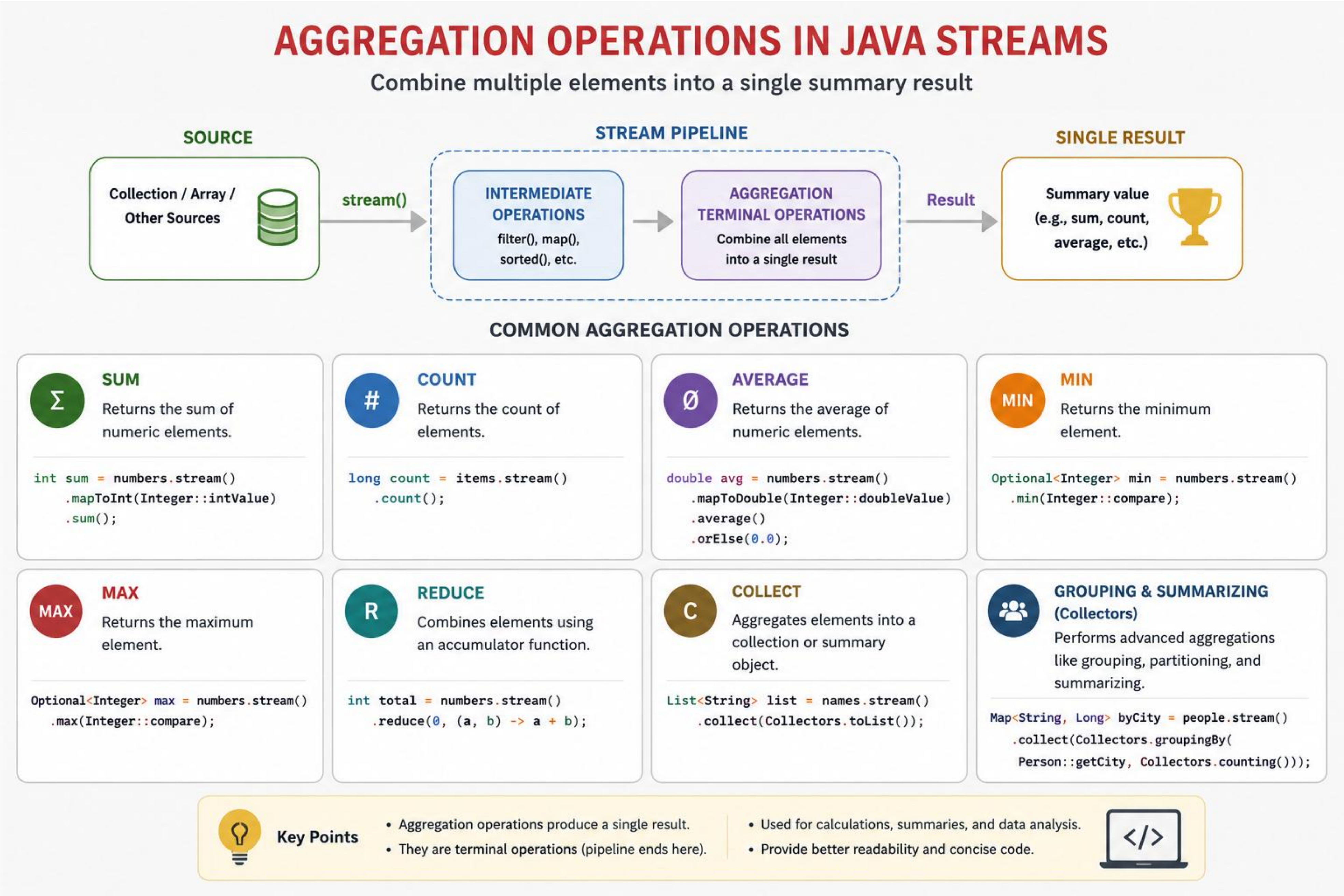
KEY TAKEAWAY reduce(identity, accumulator) always returns T; reduce(accumulator) returns Optional<T> to handle empty streams.

REDUCTION — BEST PRACTICES

Five guidelines for choosing and writing reductions correctly.

- Use built-in reduction methods (sum, min, max, count, average) whenever possible.
- Use reduce(identity, accumulator) when a natural identity exists (0 for sum, 1 for product, "" for concatenation).
- Use reduce(accumulator) when no identity value exists or when working with objects.
- Ensure the accumulator is associative and stateless for correct parallel execution.
- Prefer Optional<T> results to safely handle empty streams.

KEY TAKEAWAY Think in terms of identity, accumulator, and associativity — reduction operations are essential for aggregation and summarization.


AGGREGATION OPERATIONS IN JAVA STREAMS
Combine multiple elements into a single summary result
SOURCE
Collection / Array / Other Sources
stream()
STREAM PIPELINE
INTERMEDIATE OPERATIONS
filter(), map(), sorted(), etc.
AGGREGATION TERMINAL OPERATIONS
Combine all elements into a single result
Result
SINGLE RESULT
Summary value (e.g., sum, count, average, etc.)
COMMON AGGREGATION OPERATIONS
Σ
SUM
Returns the sum of numeric elements.
int sum = numbers.stream()
.mapToInt(Integer::intValue)
.sum();
#
COUNT
Returns the count of elements.
long count = items.stream()
.count();
Ø
AVERAGE
Returns the average of numeric elements.
double avg = numbers.stream()
.mapToDouble(Integer::doubleValue)
.average()
.orElse(0.0);
MIN
MIN
Returns the minimum element.
Optional<Integer> min = numbers.stream()
.min(Integer::compare);
MAX
MAX
Returns the maximum element.
Optional<Integer> max = numbers.stream()
.max(Integer::compare);
R
REDUCE
Combines elements using an accumulator function.
int total = numbers.stream()
.reduce(0, (a, b) -> a + b);
C
COLLECT
Aggregates elements into a collection or summary object.
List<String> list = names.stream()
.collect(Collectors.toList());
GROUPING & SUMMARIZING (Collectors)
Performs advanced aggregations like grouping, partitioning, and summarizing.
Map<String, Long> byCity = people.stream()
.collect(Collectors.groupingBy(
Person::getCity, Collectors.counting()));
Key Points
• Aggregation operations produce a single result.
• They are terminal operations (pipeline ends here).
• Used for calculations, summaries, and data analysis.
• Provide better readability and concise code.
</>

HOW AGGREGATION WORKS

Given the stream 2, 4, 6, 8, 10 — every metric computed in one pass.

METRIC	VALUE
`Count`	5
`Sum`	30
`Min`	2
`Average`	6.0
`Max`	10

```
IntSummaryStatistics stats =
    numbers.stream()
    .mapToInt(i -> i)
    .summaryStatistics();

stats.getCount();     // 5
stats.getSum();       // 30
stats.getMin();       // 2
stats.getAverage();   // 6.0
stats.getMax();       // 10

// Custom: a Collector-based aggregation
employees.stream().collect(
 summingDouble(Employee::getSalary)); // 225000.0
```

KEY TAKEAWAY Aggregation condenses many values into meaningful insights that help in decision making.

AGGREGATION — BEST PRACTICES

Five guidelines for accurate, efficient aggregation.

- Choose the right aggregation method for the data type.
- Use mapToInt(), mapToLong(), or mapToDouble() before sum()/average() for numeric streams.
- Handle Optional results from min(), max(), and average() safely.
- Use summaryStatistics() when multiple metrics are needed together.
- Aggregation operations are terminal — they produce a single result.

KEY TAKEAWAY Aggregation helps in analytics, reporting, and decision making — pick the right operation and return type.

TRY IT YOURSELF — EXERCISE 4: HIGHEST AND LOWEST SALARY

Reinforces: the aggregation operations you just saw.

STEP	CODE	RESULT
`Goal`	Find max and min salary with Comparator.comparingDouble	Max Diana 90000; min Evan 55000
`Code`	.max(comparingDouble(Employee::salary)).orElseThrow()	max()/min() return an Optional<Employee>
`Key concept`	orElseThrow() unwraps the Optional or fails fast if empty	Ties to the aggregation operations you just saw
`Reference`	exercises/exercise-04-minmax.md	Full starter code and steps

TRY IT NOW Complete Exercise 4 before continuing — see exercises/exercise-04-minmax.md.

COLLECTORS FRAMEWORK

A rich set of collector implementations to accumulate stream elements into result containers or summary values.

SOURCE		INTERMEDIATE		TERMINAL		RESULT
`Collection • Array`	→	`filter() • map() • sorted()`	→	`collect(Collector)`	→	`Container or summary value`

- Collectors are immutable, stateless, and thread-safe — all defined in java.util.stream.Collectors.
- Common use cases: grouping, partitioning, summarizing, joining, and collecting into collections.

KEY TAKEAWAY collect(Collector) is the terminal operation that triggers the pipeline and returns the result.

HOW COLLECTORS WORK

Collectors.toSet() on ["Alice", "Bob", "Charlie", "Bob"] — duplicates are removed.

Alice, Bob, Charlie, Bob → **Collectors .toSet()** → **{Alice, Bob, Charlie}**

```
List<Integer> list = numbers.stream().collect(Collectors.toList());   // [1,2,3,4,5]
Set<Integer> set = numbers.stream().collect(Collectors.toSet());      // {1,2,3,4,5}
String joined = numbers.stream().map(Object::toString)
    .collect(Collectors.joining(", "));   // "1, 2, 3, 4, 5"
```

KEY TAKEAWAY A collector defines how to accumulate elements, the result type, and how partial results combine in parallel.

COMMON COLLECTORS — PART 1

Collection and reduction collectors.

CATEGORY	COLLECTOR	DESCRIPTION	EXAMPLE
`Collection`	toList() / toSet()	Collects into a List / Set	collect(toList())
`Collection`	toCollection(Supplier)	Collects into a specific collection	toCollection(ArrayList::new)
`Reduction`	counting()	Counts the number of elements	collect(counting())
`Reduction`	summingInt() / averagingDouble()	Sums / averages values	summingInt(Employee::getSalary)
`Reduction`	minBy() / maxBy()	Min/max element using a Comparator	minBy(naturalOrder())

KEY TAKEAWAY count=5, sum=15, min=1, average=3.0, max=5 — that's what Collectors.summarizingInt() gives you in one call.

COMMON COLLECTORS — PART 2 & GROUPING EXAMPLE

Grouping and joining collectors, plus groupingBy() in action.

CATEGORY	COLLECTOR	DESCRIPTION	EXAMPLE
Grouping	groupingBy(...)	Groups elements by a classifier function	groupingBy(Employee::getDepartment)
Grouping	partitioningBy(...)	Partitions into two groups (true/false)	partitioningBy(e -> e.getAge()>30)
Joining	joining(delim, prefix, suffix)	Joins with delimiter, prefix, and suffix	joining(", ", "[", "]")

```
Map<String, List<Employee>> byDept = employees.stream()
    .collect(Collectors.groupingBy(Employee::getDepartment));

// Result: HR -> [Alice, Charlie]   IT -> [Bob, David]
```

KEY TAKEAWAY groupingBy() is the collector you'll reach for most in real business reporting.

TRY IT YOURSELF — EXERCISE 5: MAP 10% RAISE

Reinforces: map() and collecting a stream into a List, as just covered.

STEP	CODE	RESULT
`Goal`	Map salaries ×1.10, collect into a List<Double>	Updated salary list prints
`Code`	.map(e -> e.salary() * 1.10).toList()	A new List<Double> of raised salaries
`Key concept`	toList() is a terminal op that collects the stream into a List	Ties to the Collectors framework you just saw
`Reference`	exercises/exercise-05-map-raise.md	Full starter code and steps

TRY IT NOW Complete Exercise 5 before continuing — see exercises/exercise-05-map-raise.md.

TRY IT YOURSELF — EXERCISE 6: COUNT BY DEPARTMENT

Reinforces: groupingBy and counting, as just covered.

STEP	CODE	RESULT
`Goal`	Count employees per department with groupingBy + counting	Finance=1, HR=2, IT=2 (TreeMap: alphabetical)
`Code`	groupingBy(Employee::department, counting())	A Map<String, Long> of counts
`Key concept`	groupingBy buckets elements by a classifier function	Ties to the Collectors framework you just saw
`Reference`	exercises/exercise-06-group-count.md	Full starter code and steps

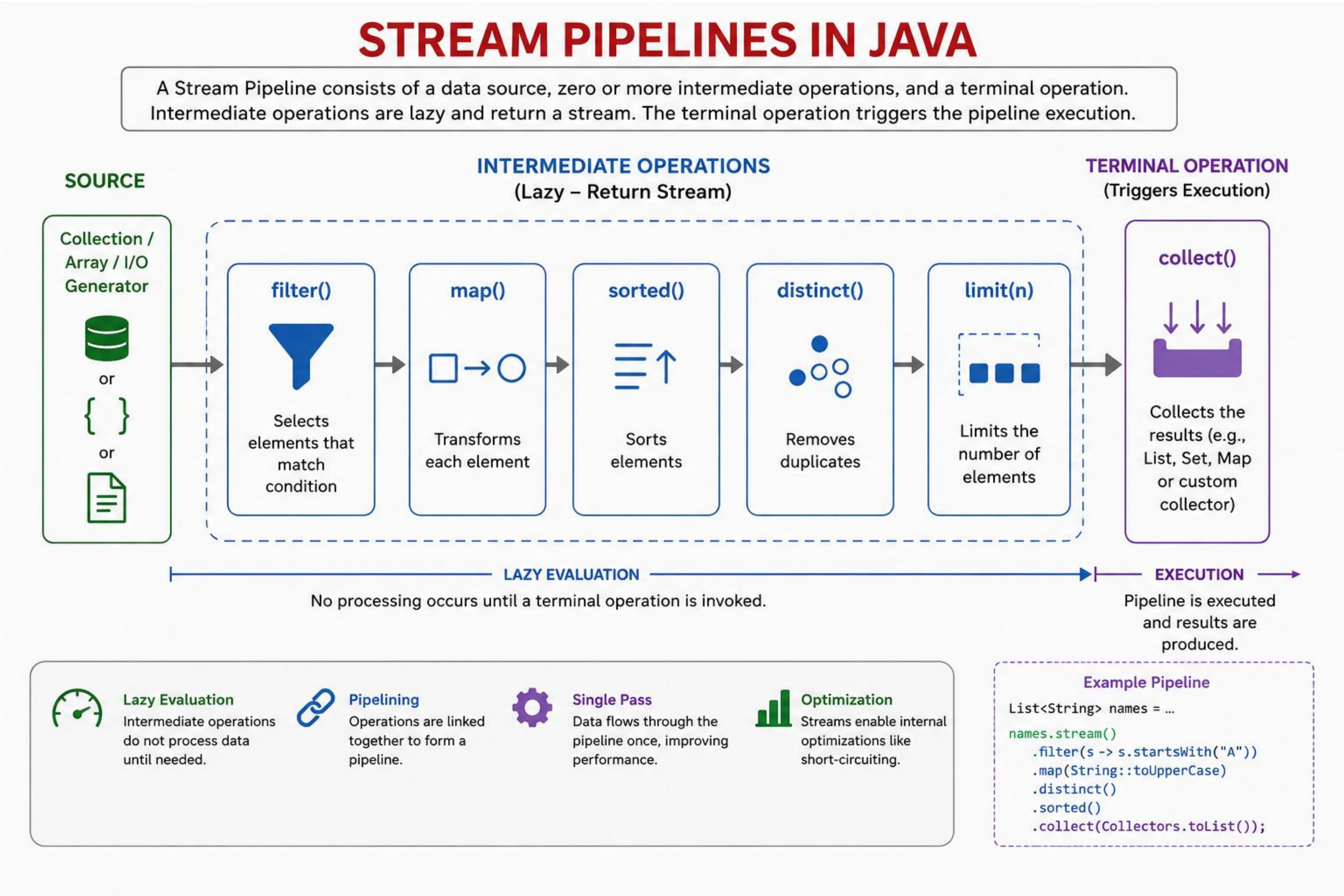
TRY IT NOW Complete Exercise 6 before continuing — see exercises/exercise-06-group-count.md.

STREAM PIPELINES

A sequence of operations on a stream: a source, zero or more intermediate operations, and a terminal operation.

STAGE	PURPOSE
`Source`	Get a stream from the data source.
`Intermediate Operation`	Transform or filter the stream (lazy).
`Intermediate Operation`	Apply additional transformations or filtering (lazy).
`Terminal Operation`	Produce a result or side effect (eager).
`Result`	Collection, value, or side effect.

KEY TAKEAWAY Pipelines are designed for readability, composability, and parallelism — and never modify the source data.


STREAM PIPELINES IN JAVA
A Stream Pipeline consists of a data source, zero or more intermediate operations, and a terminal operation.
Intermediate operations are lazy and return a stream. The terminal operation triggers the pipeline execution.
SOURCE
Collection / Array / I/O Generator
or
{ }
or
INTERMEDIATE OPERATIONS
(Lazy – Return Stream)
filter()
Selects elements that match condition
map()
Transforms each element
sorted()
Sorts elements
distinct()
Removes duplicates
limit(n)
Limits the number of elements
TERMINAL OPERATION
(Triggers Execution)
collect()
Collects the results (e.g., List, Set, Map or custom collector)
LAZY EVALUATION
No processing occurs until a terminal operation is invoked.
EXECUTION
Pipeline is executed and results are produced.
Lazy Evaluation
Intermediate operations do not process data until needed.
Pipelining
Operations are linked together to form a pipeline.
Single Pass
Data flows through the pipeline once, improving performance.
Optimization
Streams enable internal optimizations like short-circuiting.
Example Pipeline
List<String> names = …
names.stream()
.filter(s -> s.startsWith("A"))
.map(String::toUpperCase)
.distinct()
.sorted()
.collect(Collectors.toList());

INTERMEDIATE VS TERMINAL OPERATIONS

Two very different roles in the same pipeline.

Intermediate Operations

- Return another stream.
- Can be chained.
- Lazy — not executed until a terminal operation is called.
- Examples: filter(), map(), sorted(), distinct(), limit()

Terminal Operations

- Produce a result or side effect.
- End the pipeline.
- Trigger processing of all operations in the pipeline.
- Examples: forEach(), collect(), count(), reduce(), findFirst()

TERMINAL OPERATIONS BY CATEGORY

Consumption: forEach() | **Aggregation:** count(), sum(), min(), max(), reduce() | **Search:** findFirst(), findAny() | **Match:** anyMatch(), allMatch(), noneMatch() | **Materialization:** collect(), toList()

KEY TAKEAWAY Intermediate operations are lazy; terminal operations are eager — this is the single most important distinction in this module.

TRY IT YOURSELF — EXERCISE 7: HR DEPARTMENT NAMES

Reinforces: chaining filter and map into one pipeline.

STEP	CODE	RESULT
`Goal`	Get the names of employees in the HR department	Alice, Charlie
`Code`	filter(department == HR) → map(name)	filter narrows first, then map transforms
`Key concept`	Multiple intermediate ops can chain before the terminal one	Ties to the pipeline example you just saw
`Reference`	exercises/exercise-07-hr-names.md	Full starter code and steps

TRY IT NOW Complete Exercise 7 before continuing — see exercises/exercise-07-hr-names.md.

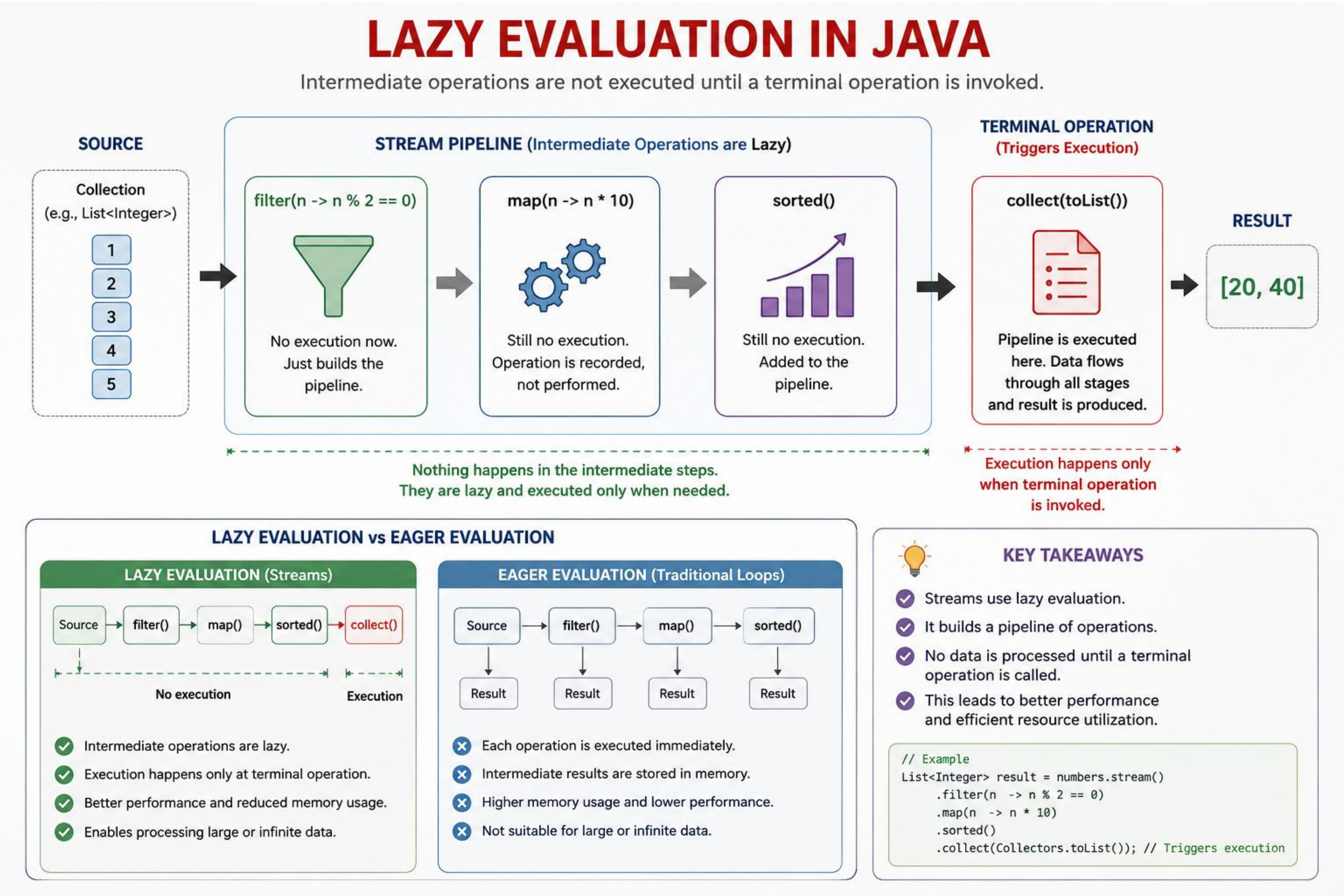
EXERCISE 7 — SOLUTION WALKTHROUGH

The full worked solution to Exercise 7 — filter, map, sort, and collect in one pipeline.

STAGE	ACTION
`Source`	employees (List<Employee>)
`filter()`	Keep only employees from the HR department
`map()`	Extract employee names
`sorted()`	Sort names in natural order
`collect()`	Collect into a List<String> (Result)

```
List<String> hrNames = employees.stream()
    .filter(e -> e.getDept().equals("HR")).map(Employee::getName)
    .sorted().collect(Collectors.toList());  // [Alice, Charlie]
```

KEY TAKEAWAY The terminal operation collect() is what triggers the entire pipeline to actually run.


LAZY EVALUATION IN JAVA
Intermediate operations are not executed until a terminal operation is invoked.
SOURCE
Collection
(e.g., List<Integer>)
1
2
3
4
5
STREAM PIPELINE (Intermediate Operations are Lazy)
filter(n -> n % 2 == 0)
No execution now. Just builds the pipeline.
map(n -> n * 10)
Still no execution. Operation is recorded, not performed.
sorted()
Still no execution. Added to the pipeline.
TERMINAL OPERATION
(Triggers Execution)
collect(toList())
Pipeline is executed here. Data flows through all stages and result is produced.
RESULT
[20, 40]
Nothing happens in the intermediate steps.
They are lazy and executed only when needed.
Execution happens only when terminal operation is invoked.
LAZY EVALUATION vs EAGER EVALUATION
LAZY EVALUATION (Streams)
Source
filter()
map()
sorted()
collect()
No execution
Execution
Intermediate operations are lazy.
Execution happens only at terminal operation.
Better performance and reduced memory usage.
Enables processing large or infinite data.
EAGER EVALUATION (Traditional Loops)
Source
filter()
map()
sorted()
Result
Result
Result
Result
Each operation is executed immediately.
Intermediate results are stored in memory.
Higher memory usage and lower performance.
Not suitable for large or infinite data.
KEY TAKEAWAYS
Streams use lazy evaluation.
It builds a pipeline of operations.
No data is processed until a terminal operation is called.
This leads to better performance and efficient resource utilization.
// Example
List<Integer> result = numbers.stream()
.filter(n -> n % 2 == 0)
.map(n -> n * 10)
.sorted()
.collect(Collectors.toList()); // Triggers execution

PIPELINE ANALOGY

Imagine a water pipeline: pipes are connected, but water doesn't flow until the tap opens.

- Pipes are connected when the pipeline is built — but nothing flows yet.
- Water (data) only flows once collect() opens the tap.

KEY TAKEAWAY Pipes are connected when the pipeline is built; water only flows when collect() opens the tap.

CODE EXAMPLE & OUTPUT

Print statements inside filter()/map() reveal exactly when each element is processed.

```
numbers.stream()
  .filter(n -> {
    System.out.println("filter: " + n);
    return n % 2 == 0;
  })
  .map(n -> {
    System.out.println("map: " + n);
    return n * 10;
  })
  .sorted();   // no terminal op yet

result.collect(Collectors.toList());
```

```
--- Building pipeline ---
--- Pipeline built, no output above ---
--- Now invoking terminal operation ---
filter: 1
filter: 2
map: 2
filter: 3
filter: 4
map: 4
filter: 5
Result: [20, 40]
```

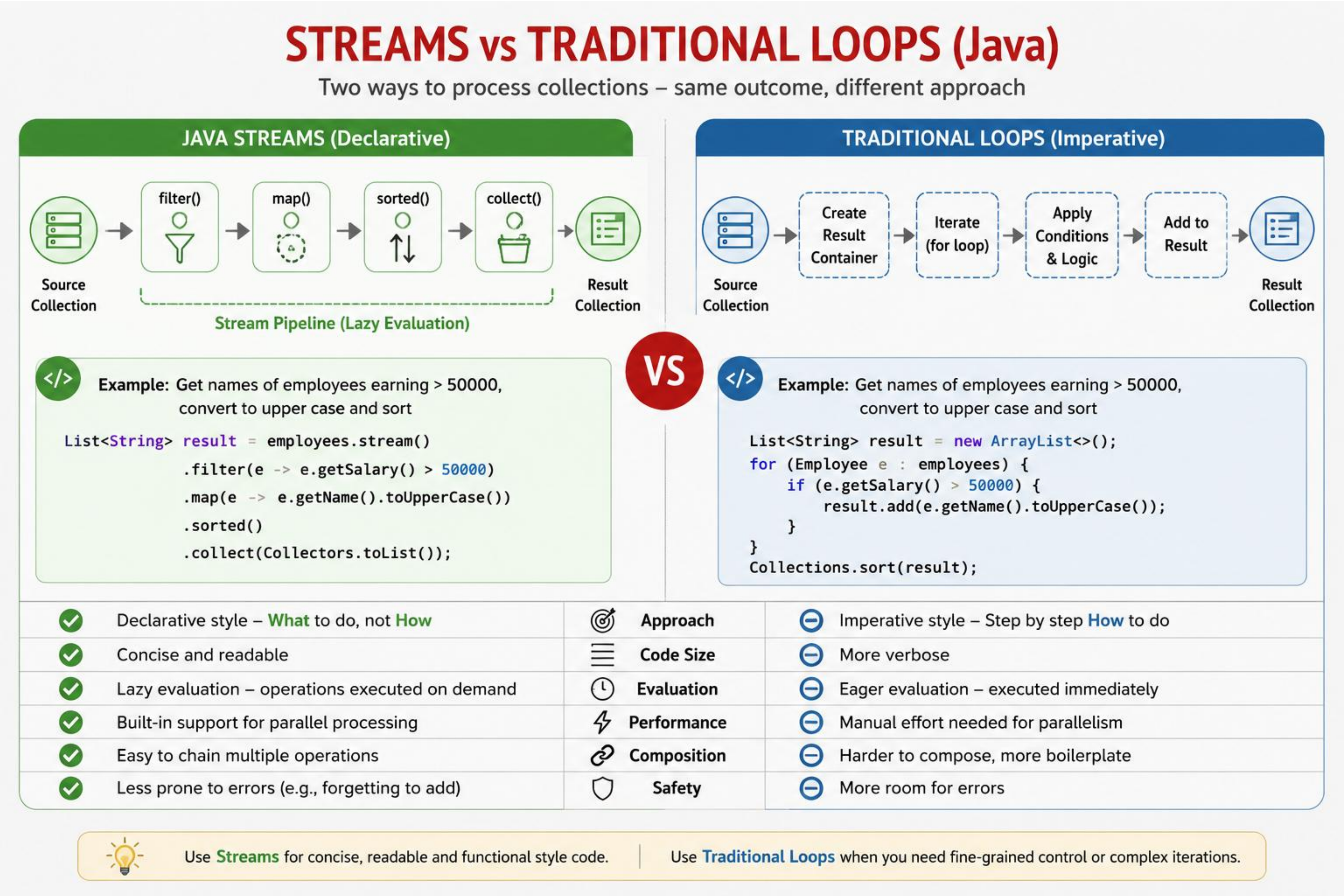
KEY TAKEAWAY Not a single print statement runs until collect() is called — proof that intermediate operations are lazy.

SHORT-CIRCUITING BENEFITS

Operations that can stop before processing the whole stream.

OPERATION	DESCRIPTION	EXAMPLE BEHAVIOR
`findFirst() / findAny()`	Returns the first / any matching element	Stops when a match is found
`anyMatch()`	Tests whether any element matches	Stops when a match is found
`allMatch() / noneMatch()`	Tests whether all / no elements match	Stops on first failure / match
`limit(n)`	Limits the number of elements	Stops after n elements
`count()`	Counts all elements	Processes the entire stream (not short-circuiting)

KEY TAKEAWAY Use short-circuiting operations when possible — they can turn an O(n) scan into an early exit.


STREAMS vs TRADITIONAL LOOPS (Java)
Two ways to process collections – same outcome, different approach
JAVA STREAMS (Declarative)
Source Collection
filter()
map()
sorted()
collect()
Result Collection
Stream Pipeline (Lazy Evaluation)
VS
Example: Get names of employees earning > 50000, convert to upper case and sort
List<String> result = employees.stream()
.filter(e -> e.getSalary() > 50000)
.map(e -> e.getName().toUpperCase())
.sorted()
.collect(Collectors.toList());
TRADITIONAL LOOPS (Imperative)
Source Collection
Create Result Container
Iterate (for loop)
Apply Conditions & Logic
Add to Result
Result Collection
Example: Get names of employees earning > 50000, convert to upper case and sort
List<String> result = new ArrayList<>();
for (Employee e : employees) {
if (e.getSalary() > 50000) {
result.add(e.getName().toUpperCase());
}
}
Collections.sort(result);
Declarative style – What to do, not How
Approach
Imperative style – Step by step How to do
Concise and readable
Code Size
More verbose
Lazy evaluation – operations executed on demand
Evaluation
Eager evaluation – executed immediately
Built-in support for parallel processing
Performance
Manual effort needed for parallelism
Easy to chain multiple operations
Composition
Harder to compose, more boilerplate
Less prone to errors (e.g., forgetting to add)
Safety
More room for errors
Use Streams for concise, readable and functional style code.
Use Traditional Loops when you need fine-grained control or complex iterations.

EXAMPLE PROBLEM

Find HR employees earning > 60,000, sort by name, and collect into a list.

ID	NAME	DEPARTMENT	SALARY
1	Alice	HR	70000
2	Bob	IT	55000
3	Charlie	HR	65000
4	David	HR	75000
5	Eve	IT	60000
6	Frank	HR	50000

Expected Output: Alice, Charlie, David

KEY TAKEAWAY David qualifies too — salary 75000 in HR — even though he's last alphabetically among the three.

SOLUTION: USING STREAMS

Concise, expressive, and easy to convert to a parallel stream.

```
List<String> result = employees.stream()
    .filter(e -> e.getDepartment().equals("HR"))
    .filter(e -> e.getSalary() > 60000)
    .map(Employee::getName)
    .sorted()
    .collect(Collectors.toList());  // [Alice, Charlie, David]
```

- Why it's better: concise and expressive; focus on what, not how; easy to read and maintain; easy to parallelize.

KEY TAKEAWAY The stream version reads almost like a description of the requirement itself.

SOLUTION: USING TRADITIONAL LOOPS

Same result, more code, and manual state management.

```
List<String> result = new ArrayList<>();
for (Employee e : employees) {
    if (e.getDepartment().equals("HR")) {
        if (e.getSalary() > 60000) {
            result.add(e.getName());
        }
    }
}
Collections.sort(result);  // [Alice, Charlie, David]
```

- Considerations: more lines of code, manual state management, more prone to errors, harder to parallelize.

KEY TAKEAWAY Nested if statements and manual sorting are exactly the kind of boilerplate streams eliminate.

WHEN TO USE WHAT?

Match the tool to the shape of the problem.

Use Streams When

- Performing transformations, filtering, mapping, sorting, or aggregations.
- You want concise, readable code.
- You may benefit from parallel processing.
- Working with large collections or I/O data.

Use Loops When

- You need fine-grained control over iteration.
- Working with complex business logic.
- You need early exit or custom control flow.
- Performance tuning simple iterations.

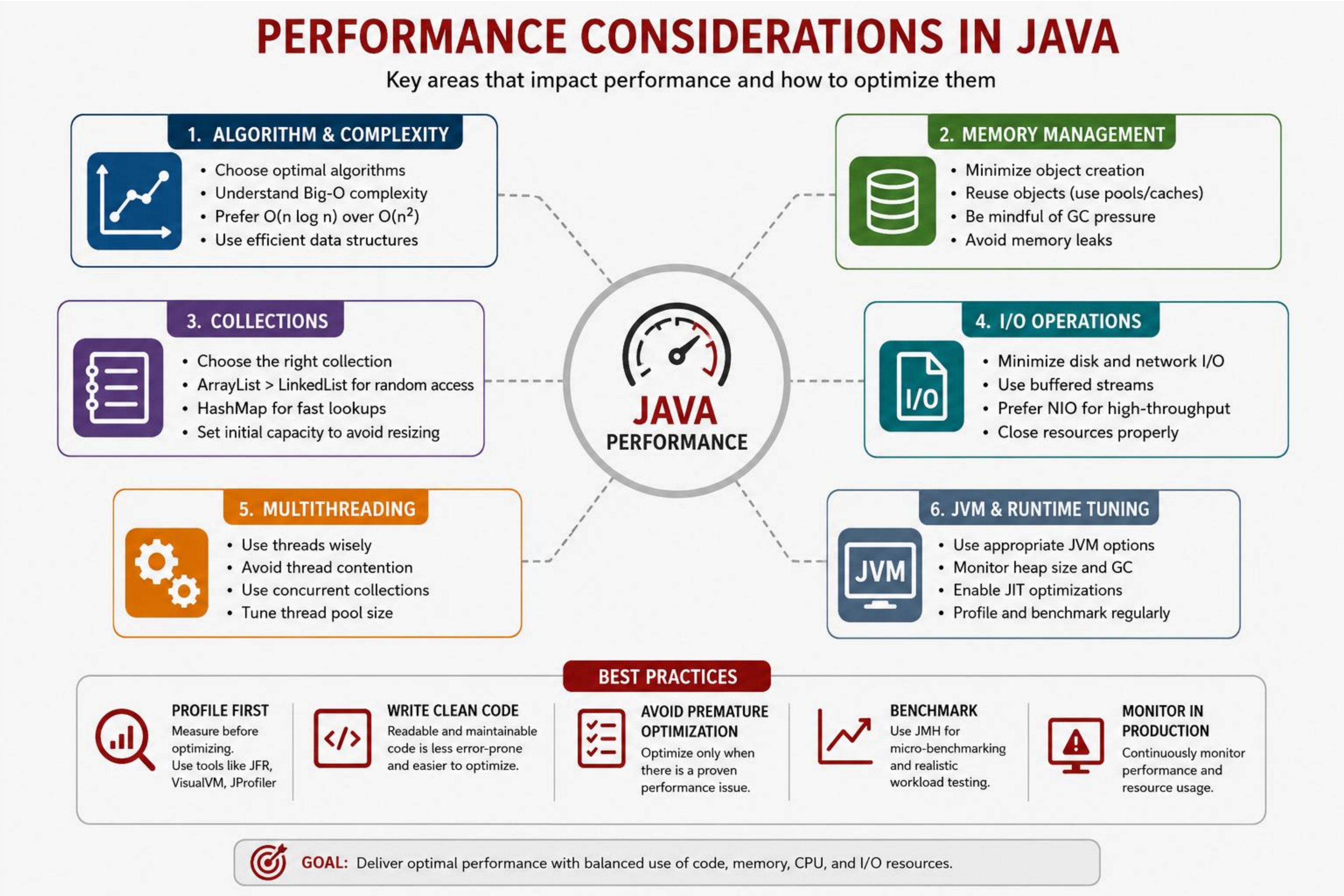
KEY TAKEAWAY Prefer streams for data processing; prefer loops for complex control flow.

PERFORMANCE CONSIDERATIONS

Streams are expressive, but understanding the trade-offs helps you write efficient, scalable applications.

- Performance depends on data size, operation type, and pipeline design.
- Use streams for readability and maintainability, but measure when performance is critical.
- Avoid unnecessary operations and boxing/unboxing.
- Be careful with stateful operations (sorted(), distinct(), limit(), etc.).
- Parallel streams can improve performance for large datasets, but may add overhead for small data.
- Always benchmark and profile before and after optimization.

KEY TAKEAWAY Streams are not always faster — measure, don't guess.


PERFORMANCE CONSIDERATIONS IN JAVA
Key areas that impact performance and how to optimize them
1. ALGORITHM & COMPLEXITY
• Choose optimal algorithms
• Understand Big-O complexity
• Prefer O(n log n) over O(n²)
• Use efficient data structures
2. MEMORY MANAGEMENT
• Minimize object creation
• Reuse objects (use pools/caches)
• Be mindful of GC pressure
• Avoid memory leaks
3. COLLECTIONS
• Choose the right collection
• ArrayList > LinkedList for random access
• HashMap for fast lookups
• Set initial capacity to avoid resizing
JAVA PERFORMANCE
4. I/O OPERATIONS
I/O
• Minimize disk and network I/O
• Use buffered streams
• Prefer NIO for high-throughput
• Close resources properly
5. MULTITHREADING
• Use threads wisely
• Avoid thread contention
• Use concurrent collections
• Tune thread pool size
6. JVM & RUNTIME TUNING
JVM
• Use appropriate JVM options
• Monitor heap size and GC
• Enable JIT optimizations
• Profile and benchmark regularly
BEST PRACTICES
PROFILE FIRST
Measure before optimizing.
Use tools like JFR, VisualVM, JProfiler
WRITE CLEAN CODE
Readable and maintainable code is less error-prone and easier to optimize.
AVOID PREMATURE OPTIMIZATION
Optimize only when there is a proven performance issue.
BENCHMARK
Use JMH for micro-benchmarking and realistic workload testing.
MONITOR IN PRODUCTION
Continuously monitor performance and resource usage.
GOAL: Deliver optimal performance with balanced use of code, memory, CPU, and I/O resources.

WHEN STREAMS PERFORM WELL — AND WHEN THEY DON'T

Two lists worth memorizing before you reach for a stream by default.

✅ Streams Perform Well When...

- Large datasets benefit from declarative processing.
- Complex transformations, filtering, grouping, aggregations.
- Leveraging parallel processing.
- Readability and maintainability matter.

⚠️ Streams May Not Be Ideal When...

- Very small datasets (stream creation overhead).
- Simple iteration with trivial logic.
- Performance-critical inner loops.
- Fine-grained control over iteration is required.

KEY TAKEAWAY The overhead of building a stream pipeline is real — it just doesn't matter until the dataset is trivially small.

OPERATION COST GUIDE

Not all stream operations cost the same.

OPERATION TYPE	EXAMPLES	COST	NOTES
`Stateless Intermediate`	filter(), map(), peek()	Low	Simple and efficient, can be fused.
`Stateful Intermediate`	sorted(), distinct(), limit(), skip()	Med–High	May require buffering or reordering.
`Short-Circuiting`	anyMatch(), findFirst(), limit()	Low–Med	Can stop early, saves work.
`Terminal (Non-Short-Circuit)`	collect(), forEach(), reduce()	Med–High	Processes all elements.
`Parallel Streams`	parallelStream()	Variable	High overhead for small data; benefits large data.

KEY TAKEAWAY Place cheap, stateless operations like filter() before expensive ones like sorted() to reduce work early.

STREAMS VS LOOPS: A REAL EXAMPLE

Sum of squares of even numbers, 0 to 1,000,000 — both approaches, same result.

USING STREAMS

```
long sum = IntStream.range(0, 1_000_000)
    .filter(i -> i % 2 == 0)
    .mapToLong(i -> (long) i * i)
    .sum();
```

USING A TRADITIONAL LOOP

```
long sum = 0L;
for (int i = 0; i < 1_000_000; i++) {
    if (i % 2 == 0) sum += (long) i * i;
}
```

- Parallel streams — Rule of Thumb: use only for large datasets (10,000+ elements) with CPU-intensive operations.
- Costs: thread management overhead, order may not be preserved, not beneficial for I/O-bound tasks.

KEY TAKEAWAY Both produce the same result — performance varies with JVM optimizations, data size, and pipeline complexity.

SAMPLE BENCHMARK & TUNING CHECKLIST

JMH results for 1 million integers — parallel only wins at scale.

APPROACH	ENVIRONMENT	AVG TIME (MS)	NOTES
`Traditional Loop`	Single Thread	28.7	Baseline
`Stream (Sequential)`	Single Thread	34.2	~19% slower (overhead)
`Stream (Parallel)`	8 cores	9.1	~3× faster for large data
`Stream (Parallel)`	8 cores, 1K data	35.8	Overhead dominates

- Is the dataset large enough to justify streams or parallel streams?
- Can I use primitive streams (IntStream, LongStream) to avoid boxing?
- Are expensive operations placed after filters? Can a short-circuiting terminal operation be used?
- Have I removed unnecessary intermediate operations, and benchmarked with representative data?

KEY TAKEAWAY Parallel streams only pay off at scale — at 1,000 elements the overhead dominates and it's actually slower.

BEST PRACTICES & COMMON PITFALLS

What to do, and the mistakes that catch even experienced developers.

Best Practices

- Avoid unnecessary boxing/unboxing — use primitive streams.
- Place filter() before expensive operations.
- Use limit(), findFirst(), anyMatch() to short-circuit.
- Measure performance with JMH, VisualVM, JProfiler.

Keep intermediate operations stateless and side-effect free; use meaningful method references/lambdas.
Prefer toUnmodifiableList/Set/Map() (Java 10+) for immutable results; streams suit large or infinite sources too.

Common Pitfalls

- Using streams for simple loops adds overhead without benefit.
- Using parallel streams for small datasets.
- Ignoring the cost of stateful operations like sorted().
- Assuming streams are always faster without measuring.

Reusing an already-consumed stream throws IllegalStateException — call .stream() fresh each time.

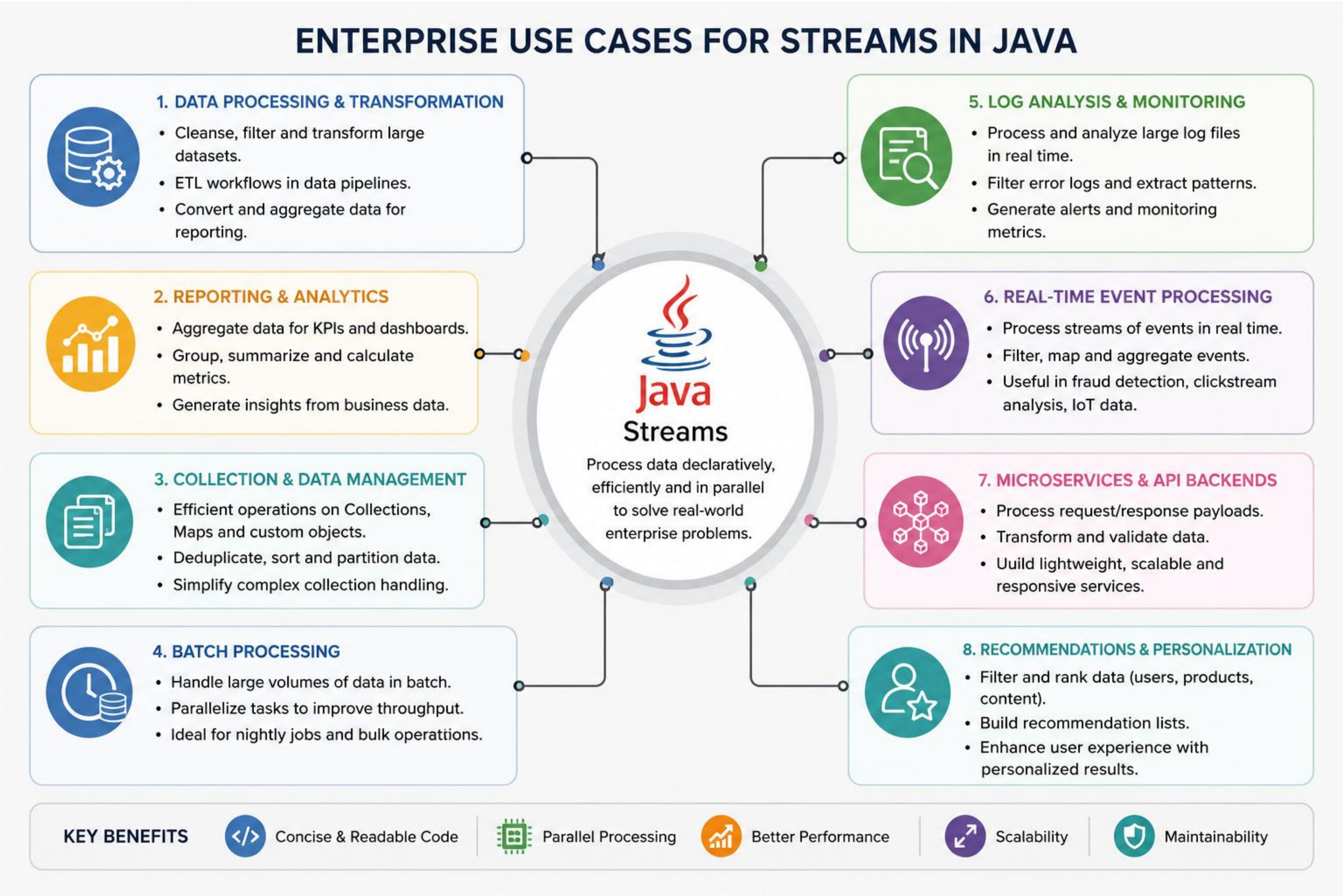
KEY TAKEAWAY Understand the trade-offs, design your pipeline wisely, and measure performance to make the right choice.

ENTERPRISE USE CASES FOR STREAMS

Streams simplify complex data processing and help build efficient, scalable enterprise applications.

- Ideal for collections, large datasets, and real-time data flows.
- Improve productivity with concise, expressive code.
- Easily support filtering, transformation, aggregation, grouping, and summarization.
- Enable parallel processing for performance at scale.
- Help build maintainable, testable, and robust applications.

KEY TAKEAWAY Streams give enterprise teams a uniform API for very different data-processing needs.


ENTERPRISE USE CASES FOR STREAMS IN JAVA
1. DATA PROCESSING & TRANSFORMATION
• Cleanse, filter and transform large datasets.
• ETL workflows in data pipelines.
• Convert and aggregate data for reporting.
2. REPORTING & ANALYTICS
• Aggregate data for KPIs and dashboards.
• Group, summarize and calculate metrics.
• Generate insights from business data.
3. COLLECTION & DATA MANAGEMENT
• Efficient operations on Collections, Maps and custom objects.
• Deduplicate, sort and partition data.
• Simplify complex collection handling.
4. BATCH PROCESSING
• Handle large volumes of data in batch.
• Parallelize tasks to improve throughput.
• Ideal for nightly jobs and bulk operations.
Java
Streams
Process data declaratively, efficiently and in parallel to solve real-world enterprise problems.
5. LOG ANALYSIS & MONITORING
• Process and analyze large log files in real time.
• Filter error logs and extract patterns.
• Generate alerts and monitoring metrics.
6. REAL-TIME EVENT PROCESSING
• Process streams of events in real time.
• Filter, map and aggregate events.
• Useful in fraud detection, clickstream analysis, IoT data.
7. MICROSERVICES & API BACKENDS
• Process request/response payloads.
• Transform and validate data.
• Uuild lightweight, scalable and responsive services.
8. RECOMMENDATIONS & PERSONALIZATION
• Filter and rank data (users, products, content).
• Build recommendation lists.
• Enhance user experience with personalized results.
KEY BENEFITS
Concise & Readable Code
Parallel Processing
Better Performance
Scalability
Maintainability

WHY STREAMS IN ENTERPRISE?

Five reasons enterprise teams standardize on the Streams API.

Higher Productivity

Less boilerplate, more focus on business logic.

Better Maintainability

Readable, modular, easy to refactor.

Performance at Scale

Lazy evaluation and parallel streams improve throughput.

Versatility

Works with collections, databases (JPA), files, real-time data.

Consistency

Uniform API for diverse data-processing needs.

KEY TAKEAWAY Consistency is underrated — the same filter/map/collect vocabulary applies everywhere in the codebase.

ENTERPRISE USE CASES — PART 1

Filtering, transformation, aggregation, and grouping.

USE CASE	DESCRIPTION	EXAMPLE OPERATIONS	BUSINESS VALUE
Data Filtering & Validation	Filter records based on business rules and quality checks	filter(), allMatch(), anyMatch()	Accurate, high-quality data
Data Transformation	Transform data into required formats or structures	map(), flatMap()	Standardizes data for processing
Aggregation & Summarization	Totals, averages, counts, min/max, summaries	collect(), reduce(), summarizingInt()	Business insights and KPIs
Grouping & Categorization	Group data by attributes, aggregate per group	groupingBy(), partitioningBy()	Helps reporting and segmentation

KEY TAKEAWAY Filtering and transformation are the entry point for almost every enterprise data pipeline.

ENTERPRISE USE CASES — PART 2

Joining, deduplication, reporting, and real-time processing.

USE CASE	DESCRIPTION	EXAMPLE OPERATIONS	BUSINESS VALUE
`Data Joining & Merging`	Combine data from multiple sources or collections	flatMap(), joining(), toMap()	Unified views across systems
`Deduplication`	Remove duplicates, ensure unique records	distinct(), toSet()	Data integrity, less redundancy
`Reporting & Dashboards`	Generate reports with aggregated data	groupingBy(), counting(), summing()	Real-time analytics for decisions
`Real-time Event Processing`	Process streaming events from queues, logs, sensors	filter(), map(), windowing	Low-latency insights and alerting

KEY TAKEAWAY Real-time event processing is where streams and message queues (Kafka, RabbitMQ) meet in production systems.

CODE EXAMPLE: EMPLOYEE ANALYTICS

Average salary per department, for employees earning more than 60,000.

```
Map<String, Double> avgSalaryByDept = employees.stream()
    .filter(e -> e.getSalary() > 60000)
    .collect(Collectors.groupingBy(
        Employee::getDepartment,
        Collectors.averagingDouble(Employee::getSalary)));
avgSalaryByDept.forEach((dept, avg) ->
    System.out.println(dept + " : " + avg));
```

DEPT	AVG SALARY
HR	74250.0
IT	68500.0
Finance	91200.0

KEY TAKEAWAY groupingBy() combined with a downstream collector is the pattern behind most enterprise analytics queries.

STREAMS IN ENTERPRISE SYSTEMS

Where streams show up when processing real production data.

- Databases (query processing) — Files & Logs (ETL, parsing, analytics)
- Message Queues (Kafka, RabbitMQ, ActiveMQ) — APIs & Services (transform/aggregate responses)
- Reports & Dashboards (analytics and visualizations)

👍 Parallel Benefits

- Utilizes multiple processor cores.
- Faster processing for large datasets.

⚠️ Considerations

- Not beneficial for small datasets.
- Order may not be preserved.
- Avoid shared mutable state.

KEY TAKEAWAY parallelStream() is best suited for large datasets and computationally intensive workloads.

PARALLEL STREAM EXAMPLE & BEST PRACTICES

Filtering high-value orders across many cores.

```
List<Customer> highValueCustomers = orders.parallelStream()
    .filter(o -> o.getAmount() > 10000)
    .map(Order::getCustomer)
    .distinct()
    .collect(Collectors.toList());
```

- Prefer streams for complex data processing; use meaningful method references and lambdas.
- Avoid side effects in intermediate operations; always handle null values appropriately.
- Consider parallel streams only after profiling and testing; measure performance in production.

Use parallel streams for large datasets (10,000+ elements) with CPU-heavy work; avoid them for small collections or I/O-bound tasks, where overhead often makes them slower.

KEY TAKEAWAY Choose streams for clarity and productivity; choose parallel streams for scale — but only after measuring, since split/merge overhead can make them slower on small data.

TRY IT YOURSELF — EXERCISE 8: PARALLELSTREAM BONUS

Reinforces: the parallel stream example you just saw.

STEP	CODE	RESULT
`Goal`	Compare count() using stream() vs. parallelStream()	Both counts match
`Code`	.stream().count() vs .parallelStream().count()	Same result either way
`Key concept`	On tiny data, parallel overhead can outweigh any benefit	Ties to the parallel stream example you just saw
`Reference`	exercises/exercise-08-parallel-bonus.md	Full starter code and steps

TRY IT NOW Complete Exercise 8 before continuing — see exercises/exercise-08-parallel-bonus.md.

PRE-LAB EXERCISE PIPELINE SHAPE

Filter, map, sort, collect — the same shape you will reuse throughout Lab 6.

Source — List<Employee>

↓

Filter — salary > 60,000

↓

Map — extract names / transform

↓

Sort — natural order

↓

Collect — to List<String> or Map

KEY TAKEAWAY Terminal operations (collect(), forEach(), count()) are what actually trigger execution.

PRE-LAB EXERCISES: CODE STARTER (SAMPLE)

The Employee record and dataset provided for the pre-lab exercises.

```java
record Employee(
        int id, String name, String department, double salary) {

    // record — canonical accessors: id(), name(), department(), salary()
}

List<Employee> employees = List.of(
    new Employee(1, "Alice", "HR", 72_000),
    new Employee(2, "Bob", "IT", 65_000),
    new Employee(3, "Charlie", "HR", 80_000),
    new Employee(4, "Diana", "Finance", 90_000),
    new Employee(5, "Evan", "IT", 55_000)
);
```

KEY TAKEAWAY This exact 5-employee dataset drives every expected output in the pre-lab exercises — Lab 6 later uses a larger 25-employee roster.

PRE-LAB EXERCISES: EXPECTED OUTPUT (SAMPLE)

What each of the seven exercises should produce, given the starter dataset.

```
1. Anonymous: true   Lambda: true
2. [Alice, Bob, Charlie, Diana]
3. [Alice, Bob, Charlie, Diana, Evan]
4. Highest Salary: 90000.0 (Diana)   Lowest Salary: 55000.0 (Evan)
5. [79200.0, 71500.0, 88000.0, 99000.0, 60500.0]
6. {HR=2, IT=2, Finance=1}
7. [Alice, Charlie]
```

Useful operations: filter(), map(), sorted(), collect(), groupingBy(), max()/min(), Comparator.comparing()

KEY TAKEAWAY Actual output may vary if the dataset changes — but the operations you'd use stay the same.

LAB 6 OVERVIEW — EMPLOYEE ANALYTICS SYSTEM

Build a menu-driven console app that turns a 25-employee roster into management reports using Streams and Lambdas.

Lab 6 Objectives

- Use lambdas and core functional interfaces (Predicate, Function, Consumer, Supplier).
- Build stream pipelines: filter, map, sorted, distinct, limit, skip.
- Aggregate with reduce(), collectors, groupingBy(), and partitioningBy().
- Handle missing values safely with Optional and build an analytics Dashboard.

Prerequisites

- JDK 21 and IntelliJ IDEA Community (or VS Code).
- Lab 0 environment setup completed.
- Comfort with List, packages, and Lab 5 service-layer patterns.
- Module 6 pre-lab exercises completed (see previous slides).

KEY TAKEAWAY Lab 6 is the real hands-on capstone for Module 6 — a full Employee Analytics System, not just the pre-lab exercises.

LAB 6 DATASET — EMPLOYEE ANALYTICS

The Employee model used throughout Lab 6 — 25 sample employees across five departments.

FIELD	TYPE	DESCRIPTION	EXAMPLE
`employeeId`	String	Primary employee key	E001
`name`	String	Employee name	John Smith
`department`	String	IT, HR, Finance, Sales, Marketing	IT
`salary`	double	Annual salary (48,000–165,000)	165000
`active`	boolean	Employment status	true

KEY TAKEAWAY Employee also has experience and rating fields — seven in total feed the Dashboard and every report.

LAB 6 CORE TASKS — MENU-DRIVEN CONSOLE

Nine graded menu options (1–9) — build them in order before any bonus features.

- 1. Display Employees — print the full 25-employee roster via forEach().
- 2. Employees By Department — group with Collectors.groupingBy(Employee::getDepartment).
- 3. Salary Report — reduce(), summarizingDouble(), and partitioningBy() by salary threshold.
- 4. Top Performers — filter and sort employees with rating >= 4.
- 5. Highest Salary — max(Comparator.comparingDouble(...)) returned as Optional<Employee>.
- 6. Department Statistics — groupingBy() + summarizingDouble() per department.
- 7. Active Employees — filter(Employee::isActive).
- 8. Dashboard — headcount, salary stats, top performer, top 5 salaries, active/inactive counts.
- 9. Exit — clean shutdown of the Scanner and menu loop.

KEY TAKEAWAY Complete CORE menu options 1–9 first; bonus features (menu 10+) come only after CORE works.

LAB 6 — KEY STREAM OPERATIONS USED

Five categories of operations power every report in the Employee Analytics System.

CATEGORY	OPERATION	EXAMPLE
`Functional Interfaces`	Predicate, Function, Consumer, Supplier	Predicate<Employee> highEarner = e -> e.getSalary() > 100_000
`Filtering & Mapping`	filter(), map(), method references	filter(Employee::isActive).map(Employee::getName)
`Sorting & Slicing`	sorted(), distinct(), limit(), skip()	sorted(Comparator.comparingDouble(Employee::getSalary).reversed()).limit(5)
`Reduction`	reduce(), mapToDouble().sum()/average()	mapToDouble(Employee::getSalary).average()
`Collectors & Optional`	groupingBy(), partitioningBy(), summarizingDouble()	groupingBy(Employee::getDepartment, summarizingDouble(Employee::getSalary))

KEY TAKEAWAY Every menu option composes these same five categories — that judgment, not the syntax, is what transfers to CRM reporting later.

LAB 6 — SUBMISSION CHECKLIST

Everything to submit and double-check once CORE menu options 1–9 and the Dashboard are working.

- 1. Complete Java Project — Employee, EmployeeData, EmployeeService, ReportService, Main under com.academy.analytics.
- 2. Screenshots — the running menu, filters/groups, and the Dashboard (option 8).
- 3. Stream Operations Table — which operations you implemented and where.
- 4. README / LMS Notes — overview, operations used, functional interfaces, how to compile and run.
- 5. Reflection Answers — written in notes/lab6-answers.md.
- 6. Bonus (Optional) — labeled stretch challenges from menu options 10+.

7. Avoid Common Pitfalls — don't reuse a consumed stream; never call .get() on Optional (use orElse()/ifPresentOrElse()); sort before both limit() and skip(); recompile with javac -d out after every change.
8. Reflection Topic — in notes/lab6-answers.md, explain why streams postpone work until a terminal operation, and how groupingBy() will reuse for CRM reporting later.

KEY TAKEAWAY Graders recompile your sources with javac -d out — the Dashboard screenshot is the single most important piece of evidence.

LAB 6 — EVALUATION RUBRIC & DASHBOARD OUTPUT

How the 100-mark rubric is graded, and what the Dashboard (menu option 8) should print.

- Graded out of 100: Project Structure (10), Lambda Expressions (10), Functional Interfaces (10), Stream Pipeline (15).
- Filtering/Mapping/Sorting (15), Collectors & Grouping (15), Reduction & Statistics (10), Dashboard & Reports (10), Code Quality (5).

```
Employees : 25   Average Salary : 100680   Highest Salary : 165000
Departments : 5   Top Performer : John Smith (Rating 5)
Highest Paid Department : IT
Active Employees : 23   Inactive Employees : 2
```

KEY TAKEAWAY Stream Pipeline, Filtering/Mapping/Sorting, and Collectors & Grouping carry the most marks — 45 of the 100.

MODULE SUMMARY AND NEXT STEPS

Java Streams and Lambda expressions provide a powerful, declarative, and concise way to process data.

- Streams enable functional-style programming using lambda expressions.
- Pipelines are built with intermediate operations and executed by terminal operations.
- Operations are lazy, concise, and promote readability.
- Streams support powerful data processing on collections, arrays, files, and more.
- Use Streams for maintainable, testable, and scalable enterprise applications.

Benefits compound across a codebase: concise & expressive, readable & maintainable, flexible across data sources, and performant at scale via parallel processing.
You can now build filter/map/collect pipelines, use Collectors for reports, and choose sequential vs parallel streams appropriately.
Did you know? The Streams API launched in Java 8. Popular entry points: collection.stream(), Arrays.stream(array), Stream.of(values...).

KEY TAKEAWAY This module covered the core concepts, operations, and practical usage of Streams through hands-on labs.

CORE CONCEPTS COVERED

Six pillars this module built up, one at a time.

λ Lambda Expressions

- Provide concise, anonymous functions.

🔗 Stream Pipelines

- Source → Intermediate Operations → Terminal Operation.

⏳ Lazy Evaluation

- Intermediate operations execute only when needed.

🧰 Stream Operations

- Filtering, mapping, sorting, aggregation, collecting.

📦 Collectors Framework

- Flexible ways to collect and summarize results.

⚖️ Performance

- When streams help, and when loops are more appropriate.

KEY TAKEAWAY Every later topic in this module built directly on lambda expressions and the pipeline shape.

WHEN TO USE — AND AVOID — STREAMS

The decision comes down to data size, complexity, and control needs.

✅ Use Streams When You Need...

- Complex data transformations.
- Filtering, mapping, and aggregation.
- Processing large datasets.
- Parallel processing for performance.
- Improved readability and maintainability.

🚫 Consider Loops When...

- Very simple iterations (stream overhead unnecessary).
- Performance-critical tight loops.
- Small datasets where readability gains are minimal.

KEY TAKEAWAY Loops still have their place when you need full control and flexibility.

IMPORTANT OPERATIONS REFERENCE

Eight operation categories, and what each one does.

CATEGORY	COMMON OPERATIONS	PURPOSE
Filtering	filter()	Select elements matching a condition
Mapping	map()	Transform each element
Sorting	sorted()	Sort stream elements
Aggregation	collect(), reduce(), summarizing*()	Aggregate or summarize data
Grouping	groupingBy()	Group elements by key
Counting	count()	Count elements
Finding	findFirst(), anyMatch(), allMatch()	Search or match elements
Collecting	toList(), toSet(), toMap()	Collect into data structures

KEY TAKEAWAY Bookmark this table — it's the fastest lookup for choosing the right operation under time pressure.

KNOWLEDGE CHECK — MULTIPLE CHOICE (1–6)

Choose the best answer. Correct answers are marked ✓.

1. **Which of the following best describes a Stream?**
   A) A data structure that stores elements **B) A sequence of elements supporting functional-style operations ✓** C) A class used to read data from files D) A thread used for parallel processing
2. **Which operation type does not trigger execution of the stream pipeline?**
   A) filter() B) map() **C) sorted() ✓** D) collect()
3. **What is the purpose of the map() operation?**
   A) To filter elements based on a condition **B) To transform each element ✓** C) To sort elements D) To remove duplicates
4. **Which terminal operation returns the number of elements in a stream?**
   A) count() ✓ B) forEach() C) collect() D) reduce()
5. **How does lazy evaluation benefit stream pipelines?**
   A) It improves code readability **B) It avoids unnecessary computation ✓** C) It increases memory usage D) It executes operations in parallel
6. **Which method processes each element without returning a result?**
   A) map() B) filter() **C) forEach() ✓** D) reduce()

KEY TAKEAWAY Six more multiple-choice questions continue on the next slide.

KNOWLEDGE CHECK — MULTIPLE CHOICE (7–12)

Choose the best answer. Correct answers are marked ✓.

7. Which collector would you use to convert a stream to a List?

A) Collectors.toSet() **B) Collectors.toList() ✓** C) Collectors.toMap() D) Collectors.toCollection()

8. What is the correct syntax to create a stream from a List<Employee> named emps?

A) Stream<Employee> s = new Stream<>(emps); **B) Stream<Employee> s = emps.stream(); ✓** C) Stream<Employee> s = Stream.of(emps); D) Stream<Employee> s = emps.toStream();

9. Which operation is used to sort elements in natural order?

A) sorted() ✓ B) sort() C) order() D) arrange()

10. Which of the following is true about parallel streams?

A) They always improve performance. B) They maintain encounter order by default. **C) They are created using parallelStream(). ✓** D) They can be used only with primitive types.

11. Which operation combines the elements of a stream into a single value?

A) collect() **B) reduce() ✓** C) map() D) filter()

12. Which statement is true about Lambda expressions?

A) They can modify only instance variables. B) They must have a return statement. **C) They must capture only final or effectively final variables. ✓** D) They can throw checked exceptions only.

KEY TAKEAWAY 12 questions, testing the full span of this module — from stream basics to lambda variable capture.

SELF-ASSESSMENT & BONUS CHALLENGE

How confident are you with Module 6 concepts?

Confident

Ready to apply these concepts independently.

Average

Comfortable with the basics, need more practice.

Need Review

Revisit the pipeline and lazy evaluation sections.

BONUS CHALLENGE

Using List<Employee> employees: filter salary > 60,000, sort by salary descending, get the top 3, collect their names into a List<String>, and print the result.

Hint: filter() → sorted() → limit() → map() → collect()

KEY TAKEAWAY If you can chain filter → sorted → limit → map → collect from memory, you've mastered this module.

Nicely done

You can write concise, declarative pipelines with Streams and Lambdas instead of imperative loops.

Exception handling and robust error management are next in your toolkit.